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The effects of AI-assisted flipped learning on critical thinking skills, dispositions, and motivation in higher education

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Abstract

Despite growing interest in AI-enhanced instruction, the specific contribution of AI to the development of critical thinking within flipped learning environments remains underexplored. This mixed-methods study investigates the impact of an integrated AI-assisted flipped learning model on university students' critical thinking skills, dispositions, and motivation. Over 10 weeks, 70 students were assigned to either an experimental group using the Socrat.ai platform within a flipped-classroom framework or a control group receiving traditional flipped learning without AI support. Quantitative data were collected using validated scales for critical thinking skills, dispositions, and motivation; pre-test equivalence checks confirmed comparability between groups, and post-test comparisons revealed greater improvements in the experimental group across all three dimensions. Qualitative data from semi-structured interviews with nine experimental-group students yielded four themes: AI-supported questioning as a catalyst for critical thinking, enhanced motivation through personalized feedback, improved preparedness for in-class discussion, and perceived challenges of AI use. These qualitative insights contextualize the quantitative gains by identifying Socrat.ai's adaptive Socratic questioning, longitudinal memory, and criterion-referenced feedback as plausible mechanisms of change. Because both groups followed the same flipped classroom structure, the observed gains are tentatively attributed to the AI-assisted component. The findings offer one of the first controlled comparisons that isolate AI's incremental contribution within an active pedagogical framework, with important implications for educators seeking to amplify critical-thinking outcomes through AI-assisted flipped learning.

Keywords AI-assisted flipped learning, Critical thinking, Motivation, University students, Mixed methods

1 Introduction

Critical thinking is one of the most fundamental academic and professional competencies of the 21st century, playing a crucial role in developing university students' innovative, analytical, and practical problem-solving skills [47]. It enables students to

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comprehend information and actively engage in questioning, analyzing, interpreting, and inductive reasoning processes [33]. Critical thinking has been consistently identified as one of the key skills required for workforce readiness and lifelong learning in major international assessments and world reports [28, 63]. As emphasized in the 2023 Global Skills Report, critical thinking ranks among the top three most essential skills for future careers (Coursera, [24]). However, employers consistently report significant gaps in graduates' abilities to analyze complex problems and evaluate information critically, and global evidence further confirms that university students often exhibit deficiencies in critical thinking skills [4, 29, 57]. For instance, Lodhi et al., [57] reported that more than 50% of university students demonstrated poor performance in inference and inductive reasoning skills. Similar findings were reported in the International Assessment of Higher Education Learning Outcomes, indicating that many university graduates lack the critical thinking skills required for success in knowledge-based economies (OECD, [64]). Such deficiencies significantly impact students' academic achievement, problem-solving abilities, and future employability [54, 88]. These findings underscore the necessity of effective pedagogical interventions to enhance students' critical thinking competencies.

While traditional pedagogical approaches often fail to develop higher-order thinking skills, the flipped learning model has emerged as a promising instructional strategy. In this model, students acquire foundational content individually before class, while classroom time is dedicated to interactive activities that promote deeper learning and critical thinking [13]. Research indicates that flipped learning can positively influence cognitive development and student engagement [15, 52]. However, implementation challenges such as instructor workload, student motivation issues, and lack of real-time guidance have limited its effectiveness [2, 61].

Recent advancements in artificial intelligence (AI) present unprecedented opportunities to address these challenges while enhancing the pedagogical effectiveness of flipped learning. AI-powered systems can provide automated content curation, personalized learning pathways, real-time student support through intelligent chatbots, and adaptive feedback mechanisms [42, 44, 69]. These technological capabilities address traditional implementation barriers while potentially creating more sophisticated learning environments that support higher-order cognitive development. From a theoretical perspective, integrating AI into flipped learning represents a synthesis of constructivist pedagogical principles with adaptive learning technologies, creating what Siemens [72] described as "connectivist" learning networks that respond dynamically to individual student needs.

Despite the theoretical potential of combining AI technologies with flipped learning, a critical gap remains in the literature: existing studies have not adequately isolated whether AI-assisted flipped learning enhances critical thinking beyond what the flipped learning structure alone provides. Most comparative designs have pitted AI-enhanced instruction against passive traditional teaching, making it impossible to determine whether observed improvements stem from the AI component, the active flipped pedagogy, or their interaction. This ambiguity has significant implications for both research design and practical decision-making. The present study was designed to close this gap through a methodological refinement described in the Research Gap and Study Objectives section below.

The findings of this study will provide valuable insights for instructors in course planning and instructional strategies, clarifying which technological features contribute most significantly to critical thinking development. Additionally, they will offer practical recommendations to educational policymakers and institutional administrators regarding digital transformation processes, contributing to optimizing student-centered learning environments. Furthermore, by enriching the literature on critical thinking and technology integration, this study aims to provide new theoretical perspectives for future research.

1.1 Critical thinking: conceptual framework and development

Critical thinking is the ability to systematically analyze, evaluate, synthesize, and reflect on information to make decisions and solve problems [33]. This definition emphasizes that critical thinking is a set of skills focusing on how individuals process information. However, critical thinking is not merely a cognitive process; it is also linked to individuals' willingness and habitual inclination to use these skills. In this regard, Facione [33] conceptualizes critical thinking as comprising skills and dispositions, highlighting its multidimensional nature. While skills include cognitive processes such as analysis, inference, and evaluation, dispositions encompass attitudinal factors such as open-mindedness, curiosity, and systematic thinking. In line with this multidimensional view, the present study employed the Istanbul Critical Thinking Scale (ICTS) to measure skills [6] and the Critical Thinking Disposition Scale (CTD) to assess dispositions [73]. The ICTS operationalizes critical thinking skills through three interrelated cognitive dimensions, interpretation, evaluation, and inference, while the CTD captures dispositional tendencies across Critical Openness and Reflective Skepticism.

Paul and Elder, [66] further contextualize critical thinking within a philosophical framework, defining it as "thinking about thinking to improve thinking"—a process guided by intellectual standards (clarity, accuracy, consistency) and intellectual virtues (humility, empathy, courage). This perspective emphasizes that critical thinking encompasses technical cognitive abilities and ethical dimensions related to character development. From an empirical standpoint, the ICTS reflects this cognitive-technical emphasis by assessing how effectively students can interpret, evaluate, and infer from given information, whereas the CTD addresses the attitudinal and ethical orientation toward engaging in critical inquiry.

Complementing these cognitive and dispositional elements, Valenzuela et al., [84] examine critical thinking motivation within the Expectancy-Value Theory framework, positing that individuals' engagement in critical thinking activities depends on both their belief in their ability to perform these activities successfully (expectancy) and the value they attribute to these activities. Accordingly, the present study employed the Critical Thinking Motivation Scale (CTM), which evaluates expectancy beliefs alongside value-related components such as attainment, utility, intrinsic interest, and perceived cost [83]. This approach allows for a more comprehensive assessment of why students engage, or fail to engage, in critical thinking processes, extending the framework beyond skills and dispositions.

Research demonstrates that critical thinking skills can be developed through specific educational interventions. Meta-analyses by Abrami et al., [1] and Huber and Kuncel [43] show that explicit instruction in critical thinking, combined with opportunities

for application and feedback, yields significant improvements in students' analytical abilities. The development of critical thinking requires educational environments that actively engage students in questioning, analyzing, and evaluating information rather than passively receiving content [47]. By integrating skills (ICTS), dispositions (CTD), and motivation (CTM), the present study adopts a multidimensional framework that reflects both the theoretical complexity of critical thinking and its practical assessment through validated instruments.

It should be noted that skills, dispositions, and motivation are treated here as interconnected rather than wholly independent constructs. From an argumentation-theory perspective, critical thinking is fundamentally a justificatory practice: individuals construct, evaluate, and revise claims through reasoned support, counter-argument, and qualification [80], and the strength of this argumentative practice is shaped by learners' epistemic beliefs [10]. In this framework, skills supply the cognitive operations required for argumentation (interpreting evidence, generating and evaluating counter-claims, drawing warranted inferences); dispositions determine the learner's willingness to engage these operations rather than defaulting to unexamined acceptance; and motivation governs the persistence with which engagement is sustained over time. The present study's three measures can therefore be read as complementary facets of a single argumentative competence rather than as three unrelated outcomes. It is also worth distinguishing the cognitive operations involved: Guilford's [39] classic distinction between divergent thinking (generating multiple candidate interpretations) and convergent thinking (narrowing these to a single, defensible conclusion) maps loosely onto the ICTS's interpretation/evaluation and inference dimensions, respectively, and AI-mediated Socratic questioning may differentially stimulate one process over the other—a possibility we return to in the Discussion. Finally, what constitutes adequate critical thinking is not discipline-neutral: evidentiary standards and argumentative norms differ across academic fields [59], and because the present sample was drawn from an elective critical-thinking course in a Faculty of Education, the findings should not be assumed to generalize to disciplines with different epistemic cultures (e.g., the natural sciences).

1.2 Flipped learning and AI integration: addressing implementation challenges

Flipped learning represents a pedagogical approach in which traditional content delivery is shifted to pre-class activities, while classroom time is dedicated to interactive learning experiences promoting higher-order thinking [12]. Grounded in constructivist learning theory, this model aligns lower-order cognitive tasks (remembering, understanding) with pre-class activities while emphasizing higher-order tasks (analyzing, evaluating, creating) during in-class sessions [5]. Research demonstrates that flipped learning can enhance student engagement and academic performance, with 93% of studies reporting improvements in at least one dimension of student engagement [17].

However, traditional flipped learning implementation faces significant challenges that limit its effectiveness. These include increased instructor workload in content preparation, student motivation issues for pre-class activities, lack of real-time guidance during independent study, technological resource limitations, and resistance to paradigm shifts [2, 61, 79]. Research indicates that these barriers can significantly impact the model's potential for fostering higher-order thinking skills.

Implementing the flipped learning model presents multifaceted challenges that can significantly undermine its effectiveness. One of the primary difficulties is the substantially increased time demand and workload placed on instructors. Studies by Hao et al. [40], Torio [79], and Zou et al. [89] indicate that instructors often struggle to find suitable video and online resources, and producing their materials is highly time-consuming. This time burden is further compounded by the need for continuous content updates and mastering video production tools. As a result, instructors may focus on preparing only specific sections of the course rather than the entire curriculum or seek assistance from teaching assistants [20, 71].

Student motivation and engagement represent a further challenge: low participation in pre-class activities and resistance to self-directed learning can undermine the model's effectiveness [8, 61]. The absence of real-time guidance outside the classroom compounds this issue, as unanswered pre-class questions create knowledge gaps that hinder in-class discussion [37, 55]. Content quality is also critical: excessively long or poorly produced videos reduce engagement, with research recommending a maximum of 5–10 min per video [14, 60]. Finally, technological resource limitations and resistance to paradigm shifts remain barriers to effective implementation [2, 7, 17]. Addressing these challenges systematically is crucial to realising the full potential of flipped learning. The following section considers how AI may provide targeted solutions to each of these barriers.

Recent technological advancements in artificial intelligence (AI) have been proposed as potential solutions to address these implementation challenges. AI-driven content curation systems can reduce instructor workload through automated video editing and intelligent content recommendation algorithms [41], functioning as AI teaching assistants that analyze existing digital resources and assist instructors in content curation. AI-supported gamification and personalized feedback systems may enhance student motivation for pre-class activities by continuously monitoring performance and providing tailored reward systems [86]. Intelligent chatbots and virtual assistants can provide real-time guidance during independent study phases, potentially addressing the critical gap in outside-classroom support [3, 44]. Additionally, AI-based video analysis tools can optimize instructional materials by assessing student engagement levels and adjusting video content structure to prevent cognitive overload [8, 50].

From a theoretical perspective, the potential integration of AI into flipped learning represents a fundamental transformation in educational practice that transcends mere task automation. The integration embodies what Kaplan and Haenlein [48] describe as "cognitive partnerships" between humans and AI, where technology serves as cognitive scaffolding that supports higher-order thinking development. This approach aligns with Clark's (2005) concept of "extended mind" tools that expand students' intellectual capacities through personalized cognitive support. Furthermore, the connection between AI-assisted flipped learning and critical thinking development is theoretically grounded in Siemens' (2005) and Downes' [30] connectivism framework, which positions AI as an adaptive node in students' learning networks. In this framework, AI provides personalized guidance that helps students navigate complex information environments and develop evaluative skills essential for critical thinking [58].

These three theoretical lenses, constructivism, connectivism, and Expectancy-Value Theory, are not merely juxtaposed in the present study but are intended to operate at

different levels of a single explanatory model. Constructivism [85] explains the proximal mechanism: Socrat.ai's Socratic Dialogue and Built-In Memory functions operate as a form of dynamic scaffolding, withdrawing or adjusting support as a student's reasoning develops, consistent with research on AI-mediated scaffolding for self-regulated learning (SRL) and, in the in-class collaborative phase, socially shared regulation of learning (SSRL), whereby groups jointly monitor and adjust their reasoning strategies [45]. Connectivism [72] explains the distal, network-level mechanism: the AI functions as one node among instructors, peers, and digital resources, and the longitudinal-memory feature extends this network across time rather than only across actors. Expectancy-Value Theory [83] explains the motivational mechanism linking the first two: scaffolded success (constructivism) raises expectancy beliefs, and a richer, more connected information environment (connectivism) increases the perceived utility and intrinsic value of the effort of critical thinking. We also recognise that effective use of an AI dialogic partner presupposes a degree of AI literacy, the ability to critically evaluate, question, and appropriately calibrate trust in AI-generated content [11], which we did not measure and which may have moderated how fully students benefited from Socrat.ai's feedback; we address this as a measurement gap in the Limitations.

1.3 Research gap and study objectives

Despite the theoretical potential of combining AI technologies with flipped learning (AAFL), research has not yet established whether the integration of AI specifically enhances critical thinking beyond what flipped learning methodology alone provides. Current studies have primarily focused on either flipped learning effectiveness in general [13, 17] or standalone AI-powered learning systems [21, 41, 44], with limited empirical investigation into the distinct and combined contributions of these approaches to critical thinking development. This represents a crucial distinction in research design, as attributing improvements to "AI-assisted flipped learning" without a control group experiencing traditional flipped learning creates ambiguity regarding which intervention component drives the observed effects.

The complex nature of critical thinking, encompassing cognitive skills, dispositional tendencies, and motivational components [32, 33, 83], requires targeted investigation using appropriately structured comparison groups to understand how specific technological enhancements affect critical thinking development. While preliminary evidence suggests that both flipped learning and AI integration can enhance student outcomes individually, the specific added value of combining these approaches, and the particular mechanisms through which AI technologies contribute to critical thinking enhancement, remain empirically underexplored.

This study addresses this gap by employing a rigorous randomized design with two comparison conditions: (1) AI-assisted flipped learning and (2) traditional flipped learning. This methodological approach isolates the specific contribution of AI technologies to critical thinking development while controlling for the effects of the flipped learning model itself. By comparing these two treatment groups, the study can determine whether improvements in critical thinking skills, dispositions, and motivation result specifically from AI-enhanced features such as adaptive feedback, personalized content delivery, and intelligent support mechanisms, or whether the flipped learning methodology alone accounts for observed gains.

The findings of this study will provide valuable insights for instructors in course planning and instructional strategies, clarifying which technological features contribute most significantly to critical thinking development. Additionally, they will offer practical recommendations to educational policymakers and institutional administrators regarding digital transformation processes, contributing to optimizing student-centered learning environments. Furthermore, by enriching the literature on critical thinking and technology integration, this study aims to provide new theoretical perspectives for future research.

Based on these objectives, the following research questions are formulated:

RQ1: Does AI-Assisted Flipped Learning enhance critical thinking skills more effectively than traditional flipped learning?

RQ2: Does AI-Assisted Flipped Learning improve critical thinking dispositions more effectively than traditional flipped learning?

RQ3: Does AI-Assisted Flipped Learning increase critical thinking motivation more effectively than traditional flipped learning?

2 Method

2.1 Research design and sample

This study employed a mixed-methods design integrating a randomized pretest-posttest control-group structure with qualitative semi-structured interviews. Individual-level random assignment was used to combine the internal-validity benefits of randomization with the ecological validity of an authentic university course setting [38], and assigned participants to either an experimental group, which received instruction through the AI-assisted flipped Learning (AAFL) approach, or a control group, which received traditional flipped learning (without AI support) (Table 1). The qualitative component aimed to contextualise and explain the quantitative outcomes by exploring students' experiences with the AI-assisted learning environment.

The key distinction between groups lies in the before-class preparation phase. While both groups engage in identical in-class interactive and collaborative learning activities designed to develop critical thinking skills, the experimental group receives AI-assisted, personalized, and adaptive learning support during pre-class preparation through Socrat.ai, whereas the control group accesses the same digital materials without AI-powered enhancements. This design isolates the specific contribution of AI technologies to critical thinking development while maintaining pedagogical consistency in the in-class learning environment.

The participants were undergraduate students enrolled in the Faculty of Education at a public university in the Marmara region of Turkey. They took the elective course "Critical and Analytical Thinking" during the 2024–2025 academic year. Random assignment

Table 1 Experimental Design

Group	Pre-test	Before-class phase	In-class phase	Post-test
Experimental (AAFL)	CT Skill, CT Disposition, CT Motivation	Socrat.ai + Digital Materials (self-paced, AI-guided, personalized feedback)	Collaborative discussions, problem-solving, group work, critical thinking activities	CT Skill, CT Disposition, CT Motivation
Control (Traditional FL)	CT Skill, CT Disposition, CT Motivation	Digital Materials Only (self-paced, no AI support, no adaptive feedback)	(identical to experimental group)	CT Skill, CT Disposition, CT Motivation

was conducted using a lottery (random drawing) procedure. After screening for eligibility, each volunteering student's name was written on an identical slip of paper; the slips were placed in a container, mixed, and drawn one at a time. To ensure allocation concealment and minimize potential bias, the lottery procedure was carried out by an individual who was independent of the researcher and was not involved in participant recruitment, data collection, or outcome assessment. The first 35 names drawn were assigned to the experimental group, and the remaining 35 students were assigned to the control group. Given this individual-level random assignment, the present study is more accurately characterised as a randomized controlled trial (RCT) employing a pretest-posttest control-group design rather than a quasi-experimental design. The experimental group consisted of 23 female and 12 male students, while the control group included 25 female and 10 male students. Gender distribution was similar across the two groups, with no statistically significant differences observed ($\chi^2(1, N = 70) = 0.27, p = .61$).

In addition to the randomized design, control variables were collected to account for potential confounding factors. These included demographic information (gender) and prior academic performance (cumulative grade point average [GPA] on a 4.0 scale). The mean GPA was 2.87 (SD = 0.17) for the experimental group and 2.85 (SD = 0.18) for the control group, indicating similar academic backgrounds between the two groups ($t(68)=0.46$, $p= .65$). These variables were incorporated into sensitivity analyses to enhance the robustness of the findings and to assess whether the intervention effects remained significant after controlling for individual differences. Prior to the intervention, participants' general experience with AI tools was also assessed via the demographic questionnaire. Across the full sample ($N = 70$), more than 95% reported having used AI tools in academic work, most frequently ChatGPT (83%) and Google Gemini (71%), with substantially lower usage rates for other tools (e.g., Microsoft Copilot, Claude, Grammarly). Self-reported usage frequency spanned the full range, from rare (8.1%, $n = 6$) to intense (16.4%, $n = 11$), with the largest proportion reporting moderate use (29.9%, $n = 21$). Because participants were randomly assigned to a condition, we have no a priori reason to expect systematic differences in prior AI experience between groups; however, we did not conduct a formal between-group equivalence test on this variable, and we note this as a transparency point rather than a validated equivalence check.

The course lasted ten weeks, with each group receiving face-to-face instruction for two class hours per week. Pretests were administered before the intervention, and posttests were conducted at the end of the ten weeks to assess changes in students' critical thinking skills, dispositions, and motivation. Ethical approval for the study was obtained from the Social and Humanities Scientific Research Ethics Committee of Bursa Uludağ University (Session No: 2024-09, Decision No: 18). The study was conducted in accordance with the ethical standards and guidelines of the committee and the principles outlined in the Higher Education Institutions Scientific Research and Publication Ethics Directive. Prior to data collection, participants were informed about the purpose of the study and their rights as participants, and written informed consent was obtained from all participants. Participation in the study was voluntary, and participants were informed that they could withdraw from the study at any time without any consequences.

All interaction data generated through Socrat.ai (e.g., student questions, AI-generated prompts, and written responses) were anonymized at the point of collection, with

student identifiers replaced by participant codes, and were used solely for instructional delivery and the research purposes described in this study. These data were securely stored for the duration of the research project and deleted upon completion of data analysis, in accordance with the ethics committee's data retention guidelines. No separate opt-out mechanism was offered specifically for AI-interaction data; students who agreed to participate consented to their Socrat.ai interaction data being used for research purposes as part of the broader informed consent process, and retained the right to withdraw from the study in its entirety at any time without consequence.

2.2 AI-assisted flipped learning in the experimental group

The experimental group followed an AI-assisted flipped Learning (AAFL) model structured into two key phases: before-class and in-class applications. Figure 1 illustrates the experimental process of the AI-assisted flipped learning model, which consists of the pre-class preparation phase (Socrat.ai interactions and digital materials) and the in-class application phase (collaborative discussions and problem-solving activities).

In the before-class phase, students engaged with digital learning materials provided by the instructor, including lecture slides and theoretical documents. Additionally, they interacted with Socrat.ai (https://socrat.ai), an AI-powered educational platform that facilitates personalized and interactive learning. The instructor provided instructional prompts to Socrat.ai each week, allowing the AI system to guide students through key concepts via interactive discussions. This process enabled students to engage in self-paced learning, receive real-time AI-generated feedback, and refine their understanding of the material before class. Students dedicated approximately 1 h per week to completing Socrat.ai activities.

The in-class phase focused on collaborative learning through structured discussions and interactive problem-solving exercises designed to develop critical thinking skills. Students participated in small-group and whole-class discussions, where they critically

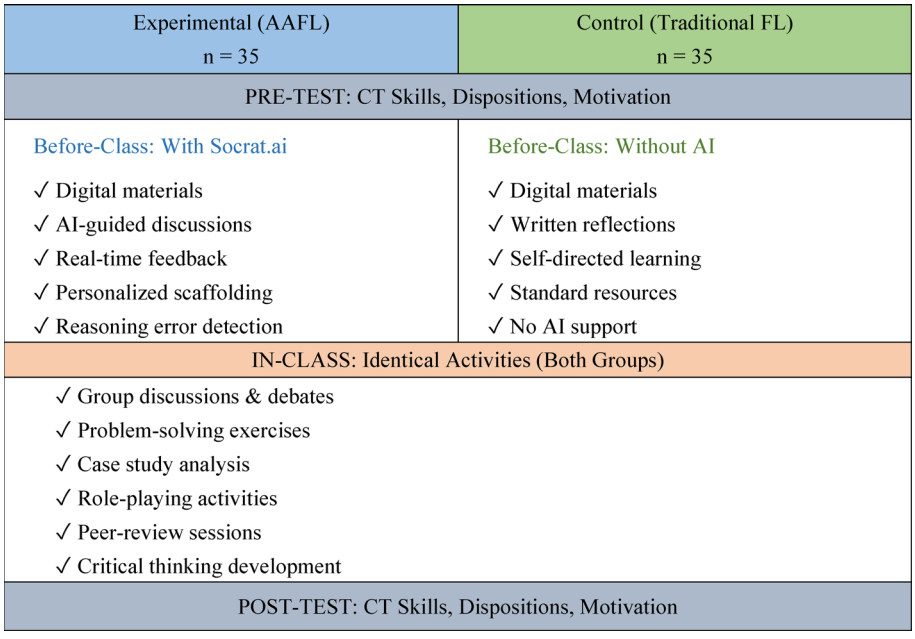


Fig. 1 Comparison of Instructional Conditions

engaged with the week's topic, shared perspectives, and refined their arguments. For example, during Week 5 (Cognitive Biases and Logical Pitfalls), students worked in small groups to analyze real-world case studies where cognitive biases affected decision-making. They then presented their findings and engaged in structured debates about potential solutions. In Week 7 (Critical Thinking and Creative Problem-Solving), students participated in collaborative problem-solving exercises. They were presented with complex scenarios requiring analytical and creative thinking, such as developing solutions for educational dilemmas using multiple perspectives and evidence-based reasoning.

During Week 9 (Critical Thinking and Ethical Decision-Making), students engaged in role-playing activities where they assumed different stakeholder positions in moral dilemmas, requiring them to construct and defend arguments from various ethical frameworks while critically evaluating opposing viewpoints. These activities included structured discourse analysis, where students examined the logical structure of their peers' arguments and provided constructive feedback on reasoning processes. The flipped learning approach ensured that students arrived at class with a foundational understanding of the subject matter, allowing classroom time to be dedicated to higher-order thinking activities, such as critical discussions and problem-solving exercises.

Week 4 (Critical Thinking Related Concepts - Logic, Reasoning, and Argumentation) also featured collaborative workshops where students constructed and deconstructed logical arguments, identifying premises, conclusions, and potential fallacies in peer-generated and published texts. Week 8 (Critical Thinking Instruction - Strategies and Methods) included reflective peer-review sessions where students critically evaluated each other's thinking processes and provided feedback on reasoning strategies.

Socrat.ai played a crucial role in individualized learning by adapting discussion prompts to students' responses, encouraging them to ask more profound questions, and providing AI-driven guidance on reasoning and argumentation. More specifically, the platform operationalized critical thinking support through three identifiable cognitive mechanisms. First, Socratic Dialogue sessions involved the AI generating open-ended, probing questions tailored to each student's prior response—for instance, when a student offered a surface-level interpretation of a cognitive bias scenario, the system would counter with a follow-up such as "What evidence would cause you to reconsider this conclusion?" thereby scaffolding the skill of reflective skepticism. Second, the platform's Built-In Memory function ensured that each new interaction was contextually linked to previous sessions, enabling the AI to detect logical inconsistencies in a student's reasoning over time and prompt metacognitive reflection (e.g., "In Week 3 you argued X; how does that position align with the evidence you presented today?"). Third, the Socrat Write feature provided structured, criterion-referenced feedback on student writing, explicitly flagging reasoning errors such as hasty generalizations or unsupported inferences and suggesting alternative argumentative pathways. These three mechanisms, Socratic questioning, longitudinal reasoning tracking, and criterion-referenced writing feedback, constitute the primary cognitive scaffolds through which Socrat.ai moved beyond content delivery to actively support the development of interpretation, evaluation, and inference skills as operationalized by the Istanbul Critical Thinking Scale (ICTS).

Because these three features were delivered as an integrated bundle rather than as separable conditions, the present design can establish the package-level contribution

of Socrat.ai but cannot determine which individual feature(s) were responsible for the observed effects; this distinction is addressed further in the Discussion and Limitations.

To illustrate how this Socratic Dialogue mechanism functioned in practice, the following exchange presents an illustrative example from pre-class preparation for Week 5 (Cognitive Biases and Logical Pitfalls), reconstructed from the platform's prompt design and the research team's observations during implementation (rather than a verbatim transcript of a specific recorded session, as session-level logs were not archived for the present manuscript). In this scenario, students were asked to evaluate a case study in which a colleague attributed a coworker's repeated lateness to laziness, without considering situational explanations:

Student: I think the coworker really is just lazy — that's the simplest explanation for being late so often.

Socrat.ai: What evidence would cause you to reconsider this conclusion?

Student: Maybe… if I found out the coworker has been dealing with something outside of work, like family issues or a long commute.

Socrat.ai: That's an interesting shift. What kind of thinking error might be at play if we jump straight to "laziness" before considering situational factors like that?

Student: I guess it could be a kind of bias where I focus too much on someone's personality and not enough on their circumstances.

Socrat.ai: How does that change your original claim? Can you restate it in a way that accounts for both dispositional and situational explanations?

This example illustrates the platform's core mechanism: rather than supplying a correct answer or labeling the bias directly, Socrat.ai used follow-up questioning to guide students toward independently identifying the relevant cognitive bias (the fundamental attribution error) and revising their initial claim, thereby preparing them to engage critically with the case-study discussion in the subsequent in-class session.

During the implementation process, several technical and practical challenges were encountered. Particularly in the first two weeks, students experienced adaptation difficulties with the Socrat.ai platform, which were resolved through individual technical support. Some students faced internet connectivity issues, causing interruptions in platform access. Alternative access methods (such as mobile application usage) were suggested to address this. Additionally, time management posed an initial challenge, as students spent more time on the platform than initially planned. To mitigate this issue, activity durations were readjusted from the second week onwards to ensure a more balanced workload.

2.3 Traditional instruction in the control group

The control group followed a traditional flipped learning approach without AI support. This group received the same digital learning materials (lecture slides and theoretical documents) as the experimental group during the before-class phase. However, instead of interacting with the AI-powered Socrat.ai platform, control group students engaged in self-paced review of these materials and completed assigned written reflection tasks on the week's topic. For example, during pre-class preparation for Week 5 (Cognitive Biases and Logical Pitfalls), control-group students received the same case-study materials as the experimental group (e.g., a written vignette describing a workplace scenario involving a possible cognitive bias) but, instead of engaging in an interactive AI dialogue,

completed a structured written reflection worksheet asking them to (a) summarise the case in their own words, (b) identify what they believed to be the central issue or potential bias at play, and (c) justify their reasoning in 150–250 words. These reflections were not graded for correctness and received no individualized feedback before the in-class session—mirroring the experimental condition's Socratic Dialogue task in topic and learning objective, but differing in interactivity, adaptivity, and immediacy of feedback. This contrast in feedback timing and individualization is itself a candidate mechanism for the AI-assisted group's advantage, alongside the differences in time-on-task discussed in the Limitations. Students had unrestricted access to standard digital resources, including library databases, course slides uploaded to the university's learning management system (LMS), and general internet resources. Importantly, they did not have access to the Socrat.ai platform or any other AI-assisted tool, and they did not benefit from personalized adaptive feedback, real-time guidance, or the AI-mediated cognitive scaffolding that characterized the experimental condition.

Crucially, the in-class phase for the control group was identical to that of the experimental group. Control group students participated in the same collaborative discussions, small-group problem-solving exercises, interactive activities, and critical thinking development tasks as the experimental group. They engaged in case study analyses, structured debates, collaborative workshops, role-playing activities, and peer-review sessions. The instructor facilitated the same types of interactive learning experiences, group discussions, and higher-order thinking activities. No gamification elements, adaptive feedback, or AI-powered mechanisms were used during the in-class sessions, but the pedagogical structure, activities, and collaborative learning environment remained consistent with the experimental group.

Course Structure and Organization

The study spanned ten weeks, with two instructional hours per week. Pre-tests were administered one week before the intervention, and post-tests were conducted one week after the final session. The experimental and control groups covered the same course topics, ensuring all students were exposed to identical content. The ten-week curriculum was structured with the following weekly topic sequence:

Week 1: The Nature of Thinking – Fundamentals and Teaching Approaches.

Week 2: Higher-Order Thinking Skills – Analysis, Synthesis, and Evaluation.

Week 3: Critical Thinking Essentials – Definitions and Misconceptions.

Week 4: Critical Thinking Related Concepts – Logic, Reasoning, and Argumentation.

Week 5: Cognitive Biases and Logical Pitfalls – Barriers to Critical Thought.

Week 6: Educational Significance of Critical Thinking – Research Evidence and Applications.

Week 7: Relationship Between Critical Thinking and Creative Problem-Solving.

Week 8: Critical Thinking Instruction – Strategies and Methods.

Week 9: Critical Thinking and Ethical Decision-Making – Moral Dilemmas and Responsible Judgment.

Week 10: Assessment of Critical Thinking Skills – Tools and Approaches.

The course aimed to develop critical thinking skills by exploring key concepts, instructional strategies, and assessment methods. Topics included the nature of thinking, higher-order thinking skills, components of critical thinking, its educational significance,

its relationship with creative problem-solving, and effective teaching and assessment methods.

The critical distinction between the two instructional conditions lay exclusively in the before-class preparation phase. While both groups engaged in identical in-class collaborative, interactive, and problem-solving activities, the experimental group benefited from AI-assisted, personalized, and adaptive learning support during pre-class preparation, whereas the control group accessed the same content materials without AI enhancement.

Each week, specific learning objectives were established and shared with both groups to ensure instructional alignment. Reading materials, case studies, and discussion prompts were identical across groups, with the primary difference being their delivery format and engagement methodology. Learning materials were carefully reviewed to ensure equal difficulty levels and content validity across the ten-week curriculum.

2.4 Socrat.ai: features and implementation in flipped learning

The primary instructional materials in this study included instructor-prepared lecture slides, digital documents, and AI-assisted activities via Socrat.ai. The instructor created a virtual classroom within Socrat.ai, designing weekly activities aligned with course objectives. The Socrat.ai teacher interface provided a comprehensive dashboard that allowed educators to create curriculum-aligned discussion topics, design interactive activities with varying difficulty levels, and analyze student performance in real time. Virtual classroom management facilitated the organization of student groups and the assignment of study materials. At the same time, an automated feedback system provided personalized responses to students. These features enabled instructors to implement AI-assisted flipped learning effectively and closely monitor student progress (Fig. 2).

Socrat.ai offered students personalized learning experiences, allowing them to engage in AI-driven discussions, receive feedback, and refine their reasoning skills.

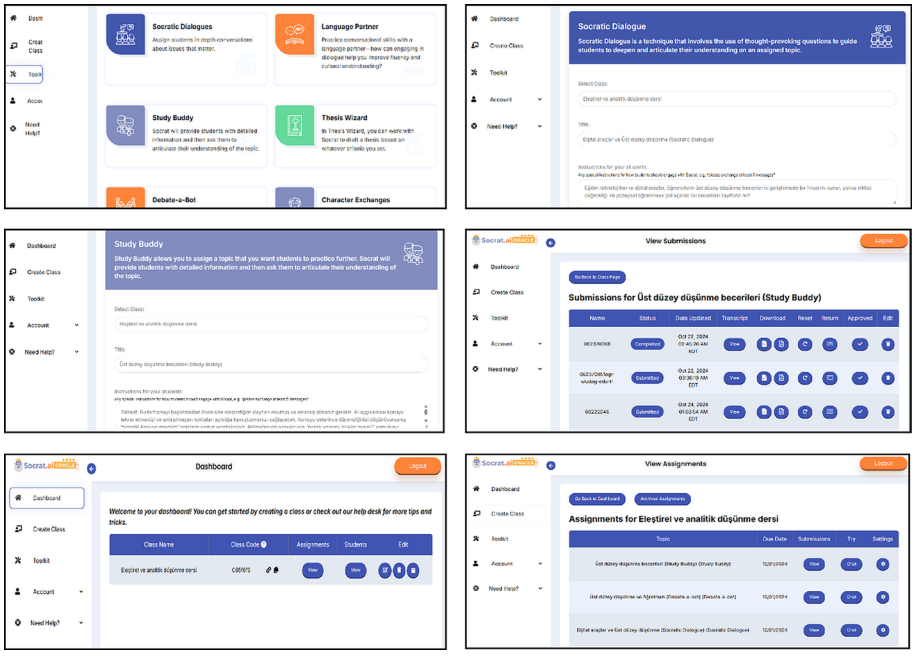


Fig. 2 Teacher Socrat.ai Accounts interface

The student interface provided an individualized and interactive environment where students accessed assigned activities, tracked their progress, and asked AI-generated questions for immediate feedback. Additionally, self-assessment tools helped students monitor their development in critical thinking skills. At the same time, collaborative workspaces supported their participation in group discussions. These features encouraged students to participate actively in pre-class preparation, ensuring they arrived at in-class sessions ready for higher-order discussions and problem-solving activities (Fig. 3).

The instructor monitored student engagement and progress through the platform's dashboard, providing targeted feedback as needed. The adaptive nature of Socrat.ai ensured that students remained continuously challenged and supported throughout the learning process.

Socrat.ai contributed to the effectiveness of flipped learning through several key features. The customizable topic feature allowed instructors to tailor pre-class materials and discussions to students' needs, increasing engagement and relevance. The Built-In Memory retained previous interactions, linking prior knowledge to new learning experiences and reinforcing metacognitive awareness. The real-time monitoring system enabled instructors to adjust instructional strategies, keeping students actively engaged. The AI-powered educational tools, including Socratic Dialogues and personalized study sessions, facilitated structured reasoning exercises and self-reflection before class.

The Socrat Play feature provided access to AI-assisted activities without requiring individual student accounts, promoting inclusivity in flipped learning. Socrat Collab facilitated small-group and whole-class discussions by summarizing key points and tracking student contributions, enriching the in-class experience. Additionally, Socrat Write supported argumentation and reasoning skills by offering personalized feedback on student writing assignments, reinforcing their ability to construct well-founded arguments. These capabilities enhanced the AI-assisted flipped Learning process, creating a more interactive, personalized, and engaging learning environment that supported students' critical thinking development.

2.5 The role of the researcher

To ensure the validity and reliability of the study, the researcher's role was carefully structured. The researcher designed the AI-Assisted Flipped Learning model and

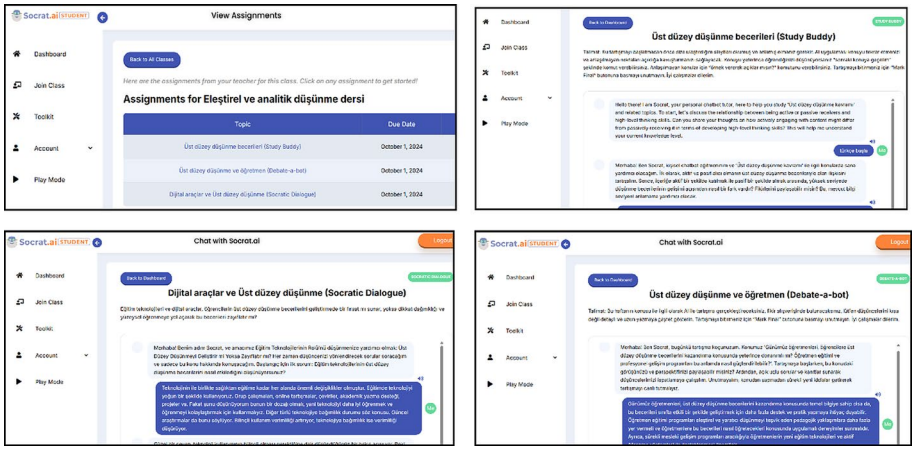


Fig. 3 Student Socrat.ai Accounts interface

developed the necessary instructional materials for the experimental group. However, following pre-defined lesson plans, an independent instructor conducted the teaching sessions to prevent potential biases.

The independent instructor was selected based on specific criteria to ensure quality instruction across both groups. These criteria included: (1) holding a Ph.D. in Education with a specialization in thinking skills, (2) having at least five years of teaching experience at the university level, (3) demonstrating proficiency in both traditional and innovative teaching methodologies, and (4) possessing no prior bias toward either instructional approach being tested. This instructor received comprehensive training on both teaching methods before the study began to ensure fidelity to the research design.

The researcher's primary responsibilities included observing student engagement, ensuring fidelity to the instructional methods, and providing technical support for using Socrat.ai. Additionally, the researcher monitored how students interacted with the AI-based system, ensuring they could effectively engage with the platform and benefit from its features. The researcher maintained detailed field notes throughout the implementation process, documenting adherence to instructional protocols and student engagement patterns. Weekly consultation meetings between the researcher and the independent instructor were held to address any implementation challenges while maintaining consistency across instructional conditions. By maintaining consistency across both instructional conditions, this approach mitigated the potential influence of the researcher on the study outcomes.

2.6 Data collection instruments

This study used three scales to measure critical thinking skills, disposition, and motivation. The details of each instrument are provided below.

2.7 Istanbul critical thinking scale (ICTS)

The ICTS, developed by Aykaç and Aslan [6], is a 25-item skill test designed for university students to measure critical thinking abilities through three cognitive dimensions: interpretation (7 items), evaluation (11 items), and inference (7 items). The test yields scores ranging from 0 to 25, with higher scores indicating stronger critical thinking abilities. The scale consists of various paragraphs and statements, each offering three response options where only one is correct. The confirmatory factor analysis (CFA) supported the three-dimensional structure of the scale, with standardized path coefficients ranging from 0.310 to 0.641, all being statistically significant. The model fit indices indicated good fit: $\chi^2/df = 1.776$, CFI = 0.966, NNFI = 0.962, GFI = 0.966, RMSEA = 0.030. The original scale demonstrated a stratified alpha coefficient of 0.676 [6]. The present study calculated Cronbach's alpha coefficient as 0.728 ($n = 140$), which exceeds the conventional minimum threshold of 0.70 but remains modest for a 25-item cognitive test. Combined with the sub-dimension reliability concerns noted below and the original validation study's CFA loadings as low as 0.310 for some items [6], this suggests that the ICTS total score, while usable, is an imperfect indicator of critical thinking skill. This measurement imprecision is more likely to attenuate than inflate the true effect of the experimental manipulation, making the present skill-level findings a conservative rather than an overstated estimate.

Because the scale's developers reported comparatively low Cronbach's alpha values for the three sub-dimensions individually (interpretation α = 0.464, evaluation α = 0.505, inference α = 0.581; [6]), and recommended using a stratified alpha for the combined total score (α = 0.676) rather than separate sub-dimension scores, the present study analyzed and interpreted only the ICTS total score. Sub-dimension-specific analyses are therefore not reported, and references to interpretation, evaluation, or inference elsewhere in this paper should be understood as illustrative of the construct rather than as evidence of differential, sub-dimension-specific effects.

2.8 Critical thinking disposition (CTD)

The Critical Thinking Disposition Scale (CTD) was developed by Sosu [73] for university students. The scale consists of 11 items and comprises two dimensions: Critical Openness and Reflective Skepticism. Critical Openness reflects individuals' tendency to actively engage with different perspectives and ideas. At the same time, Reflective Skepticism represents the disposition to question evidence and evaluate claims before accepting them. A sample item from the scale is "I often re-evaluate my experiences so that I can learn from them." The scale employs a self-report format with a 5-point Likert-type response scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). Total scores range from 11 to 55, with higher scores indicating stronger critical thinking dispositions.

Regarding the psychometric properties of CTD, Sosu [73] conducted a confirmatory factor analysis to examine the construct validity. The analysis yielded satisfactory fit indices, supporting the two-factor structure of the scale (CFI = 0.93, TLI = 0.91, RMSEA = 0.056, and SRMR = 0.055). Factor loadings ranged from 0.44 to 0.66, all statistically significant at the $p < .001$ level. The original study reported a Cronbach's alpha coefficient of 0.81, indicating good internal consistency. In the present study, the scale demonstrated acceptable reliability with an alpha coefficient of 0.78 ($n = 140$).

2.9 Critical thinking motivation (CTM)

The Critical Thinking Motivation Scale (CTM) was developed by Valenzuela et al., [83] for university students. This 19-item scale is grounded in the expectancy value theory. It aims to measure motivation toward critical thinking through two main components: expectancy and value. The expectancy component assesses individuals' competency expectations regarding critical thinking. In contrast, the value component encompasses various aspects, including attainment, utility, intrinsic value, and cost considerations. Representative items include "Concerning reasoning correctly, I am better than most of my peers" (Expectancy) and "Thinking critically will help me to be a good professional" (Utility). The scale employs a 6-point response format ranging from 1 (totally disagree) to 6 (totally agree), with total scores ranging from 19 to 114. Higher scores indicate greater motivation toward critical thinking.

Valenzuela et al., [83] employed exploratory factor analysis to evaluate the construct validity of the CTM's psychometric properties. Their findings indicated that the results aligned with the theoretical framework, confirming the expected organization of items around expectancy and value components. Specifically, factor loadings ranged from 0.328 to 0.804, explaining approximately 45% of the variance. The original study reported Cronbach's alpha coefficients ranging from 0.732 to 0.839 for the scale dimensions. In

the current study, the overall scale demonstrated good internal consistency with an alpha coefficient of 0.854 (n = 140).

2.10 Data analysis

To examine the initial differences between the experimental (AI-assisted flipped learning) and control (Traditional Flipped Learning) groups, a one-way multivariate analysis of variance (MANOVA) was performed on the pre-test scores of critical thinking skills, critical thinking dispositions, and critical thinking motivation. A one-way multivariate analysis of covariance (MANCOVA) was conducted with pre-test scores as covariates to assess the post-test differences while controlling for pre-test scores. Additionally, paired-sample t-tests were used to compare each group's pre-test and post-test scores. All statistical analyses were performed using IBM SPSS Statistics (Version 28). Qualitative data from semi-structured interviews were analysed using thematic analysis following Braun and Clarke's [18] six-phase framework; two researchers independently coded the data and member checking was conducted with three participants to enhance credibility.

To strengthen causal inferences and account for potential confounding variables, a sensitivity analysis was conducted using a multivariate analysis of covariance (MANCOVA). In this extended model, gender and cumulative grade point average (GPA) were included as additional covariates along with the pre-test scores of critical thinking skills, dispositions, and motivation. This approach allowed us to examine whether the observed effects of the instructional method remained robust after adjusting for these individual difference variables.

A priori power analysis was conducted using G*Power 3.1 [35] to determine the required sample size. Based on previous studies investigating critical thinking interventions in higher education settings [1, 62], we anticipated medium to large effect sizes (Cohen's f = 0.40) for the between-group differences [23]. With an alpha level of 0.05 and desired power of 0.80, the analysis indicated that a minimum sample size of 32 participants per group would be required to detect significant effects in an ANCOVA design with one covariate (pre-test scores). Our final sample size of 35 participants per group exceeded this minimum requirement, providing adequate statistical power while allowing for potential attrition. The sample size was also sufficient for the MANCOVA analysis. According to Tabachnick and Fidell's [78] guideline that sample size should exceed 20 times the number of dependent variables, our sample of 70 participants (with three dependent variables) met this criterion.

3 Results

3.1 Comparisons of critical thinking skills, critical thinking dispositions, and critical thinking motivation scores among the groups

A one-way between-groups multivariate analysis of variance (MANOVA) was conducted to examine whether there were significant differences between the experimental (AI-Assisted Flipped Learning) and control (Flipped Learning Only) groups in terms of their pre-test scores on critical thinking skills, critical thinking dispositions, and critical thinking motivation. Before conducting the MANOVA, preliminary assumption testing was performed, and the results indicated that all relevant assumptions were adequately satisfied, as summarized in Table 2.

Table 2 MANOVA assumption testing results

Test	Statistic	df	*p*-value	Result
Box's Test of Equality of Covariance Matrices	F = 0.485	(6, 33502.19)	0.820	Assumption met
Levene's Test – Critical Thinking Motivation	F = 1.587	(1, 68)	> 0.05	Assumption met
Levene's Test – Critical Thinking Dispositions	F = 0.116	(1, 68)	> 0.05	Assumption met
Levene's Test – Critical Thinking Skills	F = 0.405	(1, 68)	> 0.05	Assumption met
Intercorrelations Range	*r* = – .19 to – 0.10	–	–	Not highly correlated

Table 3 Comparisons of Students' Pre-Test and Post-Test CT Skill, CT Disposition, and CT Motivation Scores

Scores	Test	Group	M	SD	t(68)	*p*	Cohen's d	95% CI for Mean Diff.
Critical thinking skill	Pretest	Experimental	11.51	3.99	0.85	0.398	0.20	[– 1.04, 2.58]
		Control	10.74	3.58				
	Posttest	Experimental	15.77	4.09	2.875	0.005*	0.69	[0.80, 4.40]
		Control	13.17	3.45				
Critical thinking disposition	Pretest	Experimental	37.54	6.08	0.65	0.519	0.15	[– 2.02, 3.96]
		Control	36.57	6.45				
	Posttest	Experimental	43.00	5.48	2.541	0.013*	0.61	[0.79, 6.58]
		Control	39.31	6.61				
Critical thinking motivation	Pretest	Experimental	78.91	13.64	0.20	0.845	0.05	[– 5.49, 6.69]
		Control	78.31	11.85				
	Posttest	Experimental	94.54	11.47	3.189	0.002*	0.76	[3.26, 14.17]
		Control	85.83	11.39				

n = 35 for each group. * $p < .05$. 95% CI = 95% confidence interval for the between-group mean difference (Experimental – Control), based on pooled-variance independent-samples t-tests

An initial review of the dataset revealed that the previously reported intercorrelation values (r = .83–0.87) corresponded to each scale's own pre-post (test-retest) correlation rather than the inter-scale correlations relevant to the MANOVA assumption check; the corrected inter-scale correlations are reported above (r = – .19 to – 0.10, all non-significant, p = .12–0.41) and confirm that critical thinking skills, dispositions, and motivation are empirically distinct.

The MANOVA results revealed no statistically significant differences between the experimental and control groups in terms of their pre-test scores on critical thinking skills, critical thinking dispositions, and critical thinking motivation, Wilks' $\lambda = 0.977$, $F(3, 66) = 0.514$, p = .674, partial η^2 = 0.023 (see Table 3). These findings confirm that both groups had equivalent baseline levels across all measures. As illustrated in Fig. 4, the pre-test equivalence and the post-test improvements across groups are visually presented. It should be noted that the post-test comparisons reported in Table 3 are unadjusted (raw) group differences; the pre-test-adjusted analysis, which constitutes the primary confirmatory test for the study's central hypothesis, is reported in the following section.

As presented in Table 3, the independent samples t-tests for post-test scores revealed statistically significant differences between the experimental and control groups in all three critical thinking measures. The experimental group (AI-Assisted Flipped Learning) demonstrated significantly higher post-test scores than the control group (Flipped Learning Only) in critical thinking skills ($t(68) = 2.875$, p = .005), critical thinking dispositions ($t(68) = 2.541$, p = .013), and critical thinking motivation ($t(68) = 3.189$, p = .002).

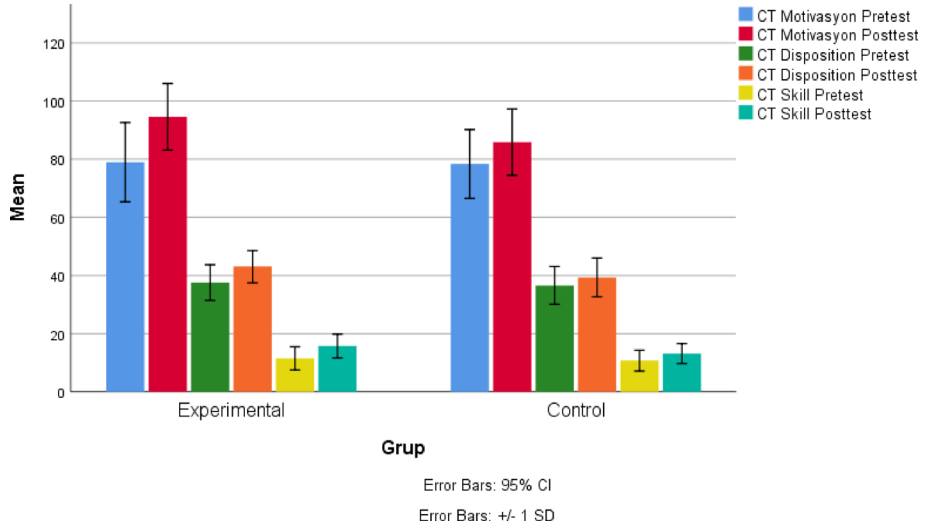


Fig. 4 Students' mean scores in the pre-test and the post-test. Error bars represent ± 1 standard deviation (SD) as reported in Table 3

Table 4 Post-test MANCOVA Assumption Testing Results

Test	Statistic	df	*p*-value	Result
Box's Test of Equality of Covariance Matrices	$\chi^2 = 4.645$	6	0.590	Assumption met
Levene's Test – CT Skill (post-test)	F = 0.891	(1, 68)	0.348	Assumption met
Levene's Test – CT Disposition (post-test)	F = 0.835	(1, 68)	0.364	Assumption met
Levene's Test – CT Motivation (post-test)	F = 0.367	(1, 68)	0.547	Assumption met
Homogeneity of regression slopes – CT Skill	F = 0.008	(1, 66)	0.928	Assumption met
Homogeneity of regression slopes – CT Disposition	F = 5.694	(1, 66)	0.020	Assumption violated
Homogeneity of regression slopes – CT Motivation	F = 2.792	(1, 66)	0.099	Assumption met

These unadjusted comparisons are descriptive; the pre-test-adjusted (ANCOVA/MANCOVA) results below constitute the primary confirmatory analysis.

To directly test the study's primary hypothesis, a one-way multivariate analysis of covariance (MANCOVA) was conducted on post-test scores of critical thinking skills, dispositions, and motivation, with the corresponding pre-test scores entered as covariates. Preliminary assumption testing (Table 4) indicated that all assumptions were satisfied, except the homogeneity-of-regression-slopes assumption for critical thinking disposition, which was violated, $F(1, 66) = 5.69$, $p = .020$; this indicates that the relationship between pre-test and post-test disposition scores differed somewhat between groups, and the corresponding univariate result should therefore be interpreted with some caution.

The MANCOVA results revealed a statistically significant overall effect of instructional group on the combined post-test dependent variables, after controlling for pre-test scores, Wilks' $\lambda = 0.423$, $F(3, 63) = 28.63$, $p < .001$, partial $\eta^2 = 0.58$. This confirms that the AI-Assisted Flipped Learning group showed significantly greater gains than the Flipped Learning Only group across critical thinking skills, dispositions, and motivation, even after adjusting for baseline differences.

Follow-up univariate ANCOVAs, evaluated against a Bonferroni-corrected alpha of 0.0167, indicated that the AI-Assisted Flipped Learning group significantly outperformed the control group on all three outcomes after adjusting for pre-test scores

Table 5 Univariate ANCOVA Follow-Up Results and Adjusted Means

Outcome	F(1,67)	*p*	95% CI for Adj. Mean Diff.	Partial η^2	Adj. M (Exp)	Adj. M (Ctrl)	Adjusted Mean Diff.	Cohen's d (adj.)
CT Skill	21.48	< 0.001	[1.09, 2.74]	0.243	15.43	13.52	1.91	0.51
CT Disposition	18.40	< 0.001	[1.52, 4.17]	0.215	42.58	39.73	2.85	0.47
CT Motivation	36.28	< 0.001	[5.52, 10.98]	0.351	94.31	86.06	8.25	0.72

Bonferroni-corrected α = 0.0167 (0.05/3). Adjusted means are estimated at the grand-mean pre-test score for each scale. Cohen's d (adjusted) = adjusted mean difference / pooled post-test SD

Table 6 Sensitivity MANCOVA: Post-test Group Effect Controlling for Pre-test, Gender, and GPA

Analysis	Statistic	Value	F	df	*p*	Partial η^2
Group (Instructional Method)	Wilks' λ	0.405	29.92	(3, 61)	< 0.001	0.595
Gender	Wilks' λ	0.918	1.80	(3, 61)	0.156	0.082
GPA	Wilks' λ	0.978	0.46	(3, 61)	0.709	0.022

Table 7 Within-Group Comparison of Pre-Test and Post-Test CT Skill, CT Disposition, and CT Motivation Scores

Scores	Group	t(34)	*p*	Cohen's d
Critical thinking skill	Experimental (AI-Assisted FL)	− 12.381	0.000*	2.09
	Control (Flipped Learning Only)	− 10.117	0.000*	1.71
Critical thinking disposition	Experimental (AI-Assisted FL)	− 9.009	0.000*	1.52
	Control (Flipped Learning Only)	− 8.406	0.000*	1.42
Critical thinking motivation	Experimental (AI-Assisted FL)	− 12.171	0.000*	2.06
	Control (Flipped Learning Only)	− 9.218	0.000*	1.56

The M and SD values are shown in Table 3, $n = 35$ for each group, * $p < .001$. Cohen's d was calculated as the mean difference divided by the standard deviation of the difference scores (dz), reflecting the strong pre-post correlation within each measure (CT Skill: $r = .87–0.92$; CT Disposition: $r = .81–0.96$; CT Motivation: $r = .83–0.92$). Because this formula is sensitive to the pre-post correlation, these within-group effect sizes are not directly comparable to the between-group effect sizes reported in Tables 3 and 5

(Table 5). The pre-test-adjusted effect sizes (d = 0.47–0.72) were somewhat smaller than the unadjusted between-group effect sizes reported in Table 3 (d = 0.61–0.76), as expected once baseline differences are statistically controlled, but remained in the medium-to-large range, confirming a robust and substantively meaningful effect of the AI-assisted intervention Table 6.

The sensitivity MANCOVA confirmed that the effect of instructional group on post-test outcomes remained statistically significant and of similar magnitude after additionally controlling for gender and GPA, Wilks' λ = 0.405, F(3, 61) = 29.92, $p < .001$, partial η^2 = 0.60. Neither gender nor GPA was a significant multivariate covariate, although gender approached significance for critical thinking motivation, F(1, 65) = 3.80, $p = .055$, suggesting a possible gender-related pattern worth exploring in future research with a larger sample. Overall, these results indicate that the observed advantage of the AI-Assisted Flipped Learning approach cannot be attributed to pre-existing differences in academic performance or to the gender composition of the groups.

3.2 Within-group comparisons of pre-test and post-test measures

Paired-sample t-tests examined the within-group differences in critical thinking skills, critical thinking dispositions, and critical thinking motivation before and after the intervention (see Table 7). The results indicated that both the experimental and control

groups showed statistically significant improvements in all three measures following the intervention.

As summarized in Table 7, the effect size results indicate substantial improvements in both groups; however, the experimental group demonstrated markedly larger effect sizes across all measures, particularly in critical thinking skills and motivation. The AI-Assisted Flipped Learning approach demonstrated superior effectiveness with very large effect sizes ($d > 2.0$) for skills and motivation, compared to the Flipped Learning Only control group ($d > 1.4$). However, because these within-group d values are calculated from the strongly correlated difference scores rather than from independent samples, they should be interpreted as descriptive of each group's own pre-to-post change rather than as a direct index of the AI-assisted approach's relative advantage; the latter is more appropriately captured by the adjusted between-group effect sizes reported in Table 5. These findings suggest that, although both instructional methods contributed to meaningful improvements, the AAFL model was particularly effective in strengthening students' critical thinking abilities, underscoring the role of AI-driven personalized learning and interactive engagement in fostering more profound cognitive development.

Taken together, both the unadjusted (Table 3) and pre-test-adjusted (Table 5) effect sizes converge on a consistent conclusion: the AI-Assisted Flipped Learning approach produced medium-to-large, educationally meaningful improvements in critical thinking skills, dispositions, and motivation relative to the Flipped Learning Only condition.

3.3 Qualitative findings

Semi-structured interviews were conducted with nine volunteer participants from the experimental group (AI-Assisted Flipped Learning) following the ten-week intervention. Participants were purposively selected to represent varied gender (4 female, 5 male) and academic performance levels (GPA 2.8–3.9). Each interview lasted 22–28 min, was audio-recorded with participants' consent, and transcribed verbatim in Turkish. Data were analysed using thematic analysis following Braun and Clarke's [18] six-phase framework, with two researchers conducting independent coding to ensure reliability. Member checking was conducted with three participants to enhance credibility and ensure interpretations aligned with participant experiences.

Four major themes emerged from the analysis: (1) AI-supported questioning as a catalyst for critical thinking, (2) enhanced motivation through personalised feedback and continuity, (3) improved preparedness for in-class discussion and collaborative confidence, and (4) perceived challenges and limitations of AI use. Each theme was validated against multiple participant accounts and grounded in excerpts from the interview data.

3.4 Theme 1: AI-supported questioning as a catalyst for critical thinking

Students consistently identified Socrat.ai's Socratic questioning mechanism as the primary driver of their critical thinking development. The platform's adaptive questioning structure pushed participants beyond surface-level responses to examine assumptions, generate supporting evidence, and test reasoning against counterarguments.

> *At first, I thought I was done after answering a question. But Socrat.ai always asked differently: 'What assumptions underlie your thinking?' I had to provide evidence and different examples.*
>
> *— Student A (Female).*

> *In traditional learning, once a question is answered, that's it. Here, Socrat.ai pushes me several steps further. I started using different layers of thinking. It wasn't just about remembering; it required reasoning.*
>
> *— Student B (Male).*
>
> *Working with Socrat.ai, I realized for the first time that explaining why my answer is correct is as important as giving a correct answer. The AI constantly asks 'why?', and I start questioning myself.*
>
> *— Student C (Female).*

These accounts demonstrate that students developed what Brookfield and Preskill [19] term "habits of mind"—moving from knowledge reproduction to metacognitive awareness. The platform's Built-In Memory feature enabled Socrat.ai to identify and target recurring thinking patterns, creating adaptive intellectual challenge rather than generic feedback. Students explicitly articulated developing core critical thinking competencies: identifying assumptions, seeking evidence, generating alternative perspectives, and constructing reasoned arguments—skills central to Ennis's [31] framework of critical thinking dispositions.

3.5 Theme 2: enhanced motivation through personalised feedback and continuity

Participants consistently highlighted how immediate, individualised feedback, particularly the platform's capacity to remember and reference prior interactions across weeks, sustained their motivation during pre-class preparation. This addresses a documented challenge in traditional flipped learning: diminished engagement during asynchronous pre-class phases.

> *When I felt bored after watching a video at home, talking to Socrat.ai drew me in. It didn't just say 'right-wrong'; it asked how I misunderstood, and I discovered my own error. When motivation is low, this kind of dialogue doesn't hold me back; it actually pulls me in."*
>
> *— Student D (Male).*
>
> *What I appreciated most was the AI remembering me. It tested last week's misconception in a new context this week. It's like having a mentor who learned my weaknesses. That keeps motivation quite high.*
>
> *— Student E (Male).*
>
> *In other courses, when I get feedback on my assignment, it's usually late and very generic. Here, Socrat.ai gives me real-time feedback, and each time it provides different, personalised responses focused on my specific errors. This motivates me to study more.*
>
> *— Student F (Female).*

These narratives exemplify Self-Determination Theory's principles [27]: personalised feedback satisfied psychological needs for competence (targeted challenge), autonomy (being asked rather than told), and relatedness (being "known" by the system). Notably, students distinguished between generic adaptive feedback and truly personalised responses grounded in long-term learning history. The Built-In Memory feature transformed isolated study sessions into continuous dialogue, addressing what research identifies as a critical gap in traditional flipped learning: the absence of meaningful interaction during pre-class preparation phases.

3.6 Theme 3: improved preparedness for in-class discussion and collaborative confidence

Students reported that engaging with Socrat.ai during pre-class preparation significantly improved their confidence, intellectual readiness, and depth of contribution to in-class collaborative activities. Pre-class Socratic dialogue functioned as cognitive and emotional rehearsal for classroom discourse.

> *In previous flipped learning courses, even when I came to class, I felt unprepared. Here it's different. With Socrat.ai in pre-class prep, I questioned topics from multiple angles. In class, instead of just saying 'I watched the video,' I could say 'there could be problems from this angle too.' I participate more actively because I feel more prepared.*
>
> *— Student G (Male).*
>
> *I noticed something: when the teacher opens a discussion in class and I can present a two-sided argument, I participate more. I was having discussions like meditation with Socrat.ai. I was basically debating with myself and the AI. Then the same discussion opens in class and my foundation is already ready. I can speak without my voice shaking.*
>
> *— Student H (Female).*
>
> *After discussing a topic with Socrat.ai in pre-class, even if that topic comes up in class, I'm in a completely different position. Because I've already explored the topic thoroughly. In class, I'm not just participating; I can act like a leader.*
>
> *— Student I (Male).*

These accounts illustrate what Schön [70] terms "reflection-in-action", intellectual rehearsal before classroom performance. Pre-class dialogue with Socrat.ai created a safe space for students to iteratively explore multiple positions, test arguments, and refine reasoning before public classroom discourse. Students described reduced anxiety and increased confidence, factors research links to both deeper engagement and retention [74]. The platform's Socrat Collab feature, which summarises discussion points and tracks contributions during in-class activities, appears to have extended this effect by creating continuity between individual reasoning and collective knowledge-building.

3.7 Theme 4: perceived challenges and limitations of AI use

Despite overwhelmingly positive experiences, participants identified several challenges, primarily technical and time-management related, that warrant consideration for implementation. These challenges did not undermine the intervention's benefits but highlight the structural supports necessary for successful deployment.

> *In early weeks, there were technical delays—questions not loading, slow responses, sometimes waiting one or two minutes. Concentration breaks. But the teacher resolved it quickly with technical support.*
>
> *— Student C (Female).*
>
> *Sometimes it takes a long time. If you allocate an hour, you sometimes can't even complete two or three questions. You don't know when to stop. Time management becomes tricky. But once the teacher told me 'maximum 45 minutes,' which helped a lot.*
>
> *— Student G (Male).*

> *The interface can be a bit confusing. When I first used it, I didn't know where to click. But the teacher showed me a tutorial, and after that it was fine. Maybe newer users need clearer guidance.*
>
> *— Student F (Female).*

These challenges are relatively minor and manageable through structural supports—technical assistance, clear time guidelines, and user training. Importantly, none of the participants attributed these issues to the pedagogical model itself; all viewed them as implementation matters resolvable through instructor guidance and platform optimisation. These findings align with technology adoption research [68], which demonstrates that initial adoption friction diminishes with support and familiarity.

Beyond cataloging these four themes descriptively, it is instructive to consider how students negotiated their relationship with Socrat.ai's feedback rather than simply receiving it. Across Themes 1 and 4, students did not appear to treat the AI's questions as authoritative; rather, they described an iterative process of testing their own answers against the AI's follow-up probes, revising when a probe exposed a gap in their reasoning, but also, particularly when technical delays or unclear prompts occurred (Theme 4), learning to recognise when the system's output reflected a technical limitation rather than a substantive challenge to their thinking. This pattern suggests an emerging, rather than fully developed, capacity for critical engagement with AI output: students in this nine-person sample described checking the AI's prompts against their own reasoning, but none spontaneously raised the possibility that Socrat.ai's questions or feedback could themselves be substantively wrong or misleading, which may indicate a degree of unexamined trust in the AI's epistemic authority that warrants closer scrutiny in future research. This nuance tempers the otherwise positive thrust of the qualitative findings and should be read alongside the discussion of the risk of overreliance below.

3.7.1 Synthesis and integration with quantitative findings

Qualitative findings are consistent with, and help contextualise, the quantitative improvements observed in critical thinking skills, disposition, and motivation. The qualitative data are consistent with a synergy among Socrat.ai's three core features, adaptive Socratic questioning, long-term memory enabling personalized feedback, and real-time interaction during pre-class preparation, that together created conditions favorable for deep learning. Because these features were bundled rather than experimentally varied, we can speak confidently to the package-level effect of Socrat.ai relative to non-AI flipped learning, but any claim about which specific feature contributed most should be treated as a tentative, hypothesis-generating interpretation rather than an established finding; isolating individual feature contributions would require a dismantling study, which we note as a direction for future research. Students experienced intellectual challenge balanced with support and recognition: the conditions research identifies as essential for critical thinking development [19]. The pre-class dialogue functioned as both cognitive scaffolding (addressing foundational skills) and motivational support (sustaining engagement), explaining why students showed improvement not only in critical thinking skills but also in disposition (willingness to engage in critical thinking) and motivation. Crucially, these process-level insights offer a plausible, qualitative account of how AI-assisted flipped learning may achieve its quantitative effects, lending

interpretive depth to the statistical findings even though the bundled design precludes a definitive feature-by-feature causal account.

4 Discussion

This study aimed to examine the effects of the AI-Assisted Flipped Learning (AAFL) model on university students' critical thinking skills, dispositions, and motivation. The research was conducted with volunteer students enrolled in the "Critical and Analytical Thinking" course at a public university in the Marmara region of Turkey. A randomized pretest-posttest control-group design was employed, where the experimental group received instruction through a Socrat.ai-based AI-assisted flipped learning approach. In contrast, the control group received instruction through a standard flipped learning approach without AI support.

Data were collected using validated and reliable scales measuring critical thinking skills, dispositions, and motivation. In the analysis, one-way MANOVA was conducted to confirm pre-test group equivalence, followed by independent samples t-tests to compare post-test outcomes between the two groups. Additionally, paired-sample t-tests were conducted within each group to examine pre-to-post changes. Qualitative semi-structured interviews with nine experimental group students provided process-level insights that contextualise and explain the quantitative outcomes. The main findings indicate that students in the experimental group exhibited significantly improved cognitive processes, critical thinking dispositions, and motivational components compared to those in the control group. These results suggest that the AI component of the flipped learning model, rather than the flipped structure per se, enhances critical thinking development by facilitating personalized learning, interactive engagement, and adaptive feedback mechanisms. In this section, we interpret the key findings, offer educational policy and instructional practice recommendations, and discuss the study's limitations.

4.1 Effects on critical thinking skills

Our analysis revealed that students in the experimental group demonstrated significantly improved critical thinking skills, as measured by the ICTS total score, compared to the control group. Because sub-dimension scores were not analyzed separately (see Method), we cannot determine whether this improvement was concentrated in particular cognitive processes such as interpretation, evaluation, or inference, or distributed across all three; descriptively, the finding suggests that students moved beyond surface-level knowledge acquisition toward a more analytical engagement with information.

A key methodological strength of this study is the use of a control group that received flipped learning without AI support. This design isolates the contribution of AI assistance from the flipped learning structure itself. Both groups benefited from the same flipped classroom framework, including pre-class digital materials, in-class collaborative activities, and structured discussion; the only systematic difference was the presence or absence of the Socrat.ai AI platform in the pre-class preparation phase. Consequently, the significantly greater improvements observed in the experimental group can be attributed more confidently to the AI-assisted component rather than to the flipped learning structure alone. Although, as discussed in the Limitations, factors such as differential time-on-task during pre-class preparation cannot be fully ruled out as a contributing mechanism.

The mechanism of change can be understood at three interconnected levels. At the cognitive scaffolding level, Socrat.ai's Socratic Dialogue feature systematically targeted the three ICTS dimensions: interpretation was challenged through context-shifting prompts that required students to re-read information under different assumptions; evaluation was engaged through counter-argument generation tasks in which the AI presented opposing evidence; and inference was practiced through hypothetical scenario extensions that required students to draw conclusions beyond the given data. At the metacognitive level, the platform's Built-In Memory function created a longitudinal feedback loop, enabling students to track the evolution of their own reasoning across weeks—a process aligned with Paul and Elder's (2006) notion of thinking about thinking to improve thinking. At the motivational-regulatory level, the immediate, personalized nature of AI feedback reduced the frustration commonly associated with unanswered pre-class questions [37], thereby sustaining engagement and reinforcing students' self-efficacy beliefs as conceptualized within Expectancy-Value Theory [83].

The literature supports the role of AI-assisted instruction in cognitive development. Studies by Huang et al., [42] and Chiu et al., [21] have shown that AI enhances cognitive processes by providing personalized feedback and adaptive learning materials. Additionally, AI-powered learning environments promote student-centered learning, offering potential benefits for developing critical thinking-related skills. For instance, Suwardika et al., [77] found that an AI-supported flipped classroom model fostered creativity, digital literacy, and self-efficacy through independent learning and collaborative activities. Similarly, Huesca et al., [44] demonstrated that a ChatGPT-assisted flipped learning strategy improved student performance by creating a more interactive and personalized learning experience.

Recent empirical work reinforces the plausibility of these mechanisms while also cautioning against treating AI as a uniformly positive substitute for human guidance. Banihashem et al., [9] compared AI-generated and peer-generated feedback on argumentative essays. They found that while AI feedback was reliably structural and descriptive, human peers were better at identifying the substantive reasoning problems underlying weak arguments, suggesting that AI's contribution to critical thinking may be most effective when it complements rather than replaces human feedback. Building on this, Banihashem et al., [11] propose a "hybrid intelligent feedback" framework in which AI-generated suggestions are embedded within dialogic, reflective learning environments to support argumentation-based reasoning, rather than delivered as standalone corrections. Socrat.ai's design in the present study is broadly consistent with this hybrid logic: AI-generated Socratic prompts were embedded within an instructor-curated, peer-discussed flipped structure rather than functioning as an isolated feedback channel, which may help explain why gains were observed across skills, dispositions, and motivation rather than skills alone.

While these studies highlight the general advantages of AI-assisted learning, our design held the flipped learning structure constant across both groups, allowing for a more direct comparison. The combined approach of AI-assisted flipped learning appears to have facilitated these improvements through what we conceptualize as 'pedagogical synergy'—where the pre-class AI-guided preparation ensures students arrive with foundational knowledge and initial critical thinking practice. At the same time, in-class activities allow for collaborative refinement and application of these skills with peer and

instructor feedback. This creates a continuous cycle of skill development that the flipped structure alone, as evidenced by the control group's comparatively more modest gains, could not fully achieve.

This interpretation is further supported by findings from Kong's [51] three-year trial in Integrated Humanities, where a flipped classroom strategy incorporating online pre-lesson preparation, in-class group discussions, and after-class extended learning on a social platform was implemented. The study demonstrated that improvements in students' critical thinking, particularly in hypothesis identification, induction, and deduction, were primarily due to sufficient instructional time, structured group sharing, and explicit pedagogical guidance in critical thinking tasks, rather than the flipped format alone. Our findings extend Kong's insight by demonstrating that AI assistance is precisely the mechanism that provides this 'explicit pedagogical guidance' within a flipped framework—as evidenced by the fact that the control group, which had the same flipped structure but no AI guidance, showed significantly smaller gains.

Theoretically, our findings align with Vygotsky's social constructivism (1978), emphasizing the importance of social interaction and scaffolding in cognitive development. As Filali Marzouki et al., [36] argue, technology-enhanced learning environments, particularly those grounded in social constructivist principles, can effectively support the development of higher-order thinking skills. Our study's AI-assisted flipped learning model provided this scaffolding through personalized feedback and guided discussions, facilitating students' progression within their zones of proximal development [46]. This theoretical framework helps explain why students in the experimental group demonstrated enhanced critical thinking abilities, as they received tailored AI-driven guidance that bridged the gap between their current understanding and potential capabilities—guidance that was absent in the control group despite an otherwise identical instructional structure.

Furthermore, the flipped learning model's emphasis on in-class applications and group discussions may have reinforced this development [12, 16, 82]. Classroom activities such as brainstorming, debates, and small-group discussions gave students more opportunities to practice critical thinking. These interactive methods likely strengthened their ability to interpret, evaluate, and make inferences.

It is instructive to situate our findings within the narrower literature comparing flipped learning alone versus AI-augmented flipped learning. Katona and Gyönyörű [49] compared a standard flipped classroom with an AI-adaptive flipped condition and reported moderate gains attributable specifically to the AI component in engagement and learning outcomes, though effect sizes remained within conventionally expected ranges ($d < 1.0$). Similarly, Li [53] found that a ChatGPT-guided flipped approach produced higher project performance than a non-AI flipped condition, yet the advantage was modest. Our study employs an analogous comparison, AI-assisted flipped learning versus flipped learning only, and the between-group effect sizes observed ($d = 0.69$ for skills, $d = 0.61$ for dispositions, $d = 0.76$ for motivation) fall within the medium-to-large range, consistent with Katona and Gyönyörű's [49] findings while reflecting a more direct methodological test of AI's incremental contribution. The most parsimonious explanation for these gains is that the combination of (a) explicit critical thinking instruction embedded in the course content, (b) the shared flipped structure that freed in-class time for higher-order practice, and (c) AI-assisted scaffolding in the experimental group created

a mutually reinforcing ecology that exceeded what the flipped structure alone could produce.

However, it is worth noting that not all research has found uniformly positive effects of flipped learning on cognitive development. Some studies, such as Birgili and Demir [14], have identified challenges, including student preparation issues and technical difficulties that can impede the effectiveness of flipped learning. The success of our intervention may be partly attributed to the structured implementation and ongoing support provided to students, which helped mitigate these potential challenges. The qualitative findings illuminate the specific mechanisms through which these cognitive gains were achieved. Students described Socrat.ai's adaptive questioning as forcing deeper reasoning—moving from recall to assumption identification, evidence generation, and argument construction. As Student B articulated, the platform pushed them "several steps further" than traditional question-answer formats, requiring genuine reasoning rather than reproduction. This process-level account is broadly consistent with the overall improvement observed on the ICTS total score, although, as noted in the Method, the present design does not permit us to confirm whether these gains were concentrated in specific sub-dimensions. Testing this possibility with an instrument offering stronger sub-dimension reliability is a worthwhile direction for future research.

4.2 Effects on critical thinking disposition

Our analysis also indicated that students in the experimental group exhibited more significant improvements in critical thinking disposition than those in the control group. The Critical Thinking Disposition Scale (CTD) measures students' overall tendency to engage in critical thinking behaviors, encompassing their general willingness to engage with diverse perspectives and question claims before accepting them [73]. The findings demonstrate that students in the AI-assisted flipped learning model developed significantly higher overall critical thinking disposition scores than the flipped-learning-only control group.

Existing studies highlight that the flipped learning model enhances student interaction and strengthens critical thinking [56, 65, 87]. Additionally, AI-assisted applications are known to foster dynamic discussion environments and provide real-time feedback, which can further enhance students' inquisitive attitudes [3, 44, 75, 86]. Our study extends these findings by providing more direct evidence that it is specifically the AI component, not merely the flipped structure, that drives dispositional gains: since both groups experienced the same flipped learning environment, the significant between-group difference in disposition scores points to AI-assisted pre-class preparation as the active ingredient. The AI-driven interactions cultivate initial questioning habits and openness to multiple perspectives. In contrast, flipped classroom discussions, shared equally by both groups, provide authentic contexts for students to practice and internalize these dispositional tendencies through peer interaction and real-time application.

More specifically, Socrat.ai's personalized feedback, automated analysis, and guided discussions encouraged students to examine their thoughts critically. The AI system's consistent modeling of questioning strategies and perspective-taking during pre-class preparation appears to have established cognitive habits that students then transferred to in-class collaborative activities, creating a sustained environment for dispositional development beyond any single learning context. As students engaged in AI-driven

discussions, instant feedback from the system helped them identify inconsistencies or weak arguments, contributing to their overall disposition to engage in critical thinking processes. Additionally, AI-supported discussions exposed students to alternative viewpoints, helping them develop a more flexible and open-minded approach to knowledge acquisition and evaluation.

Our findings regarding critical thinking dispositions can be understood through Facione's (1990) conceptualization of critical thinking as comprising both skills and dispositions. The AI-assisted flipped learning model appears to nurture what Paul and Elder [66] describe as 'intellectual virtues,' such as intellectual humility, empathy, and courage. By engaging with AI-generated feedback and diverse perspectives, students likely developed greater awareness of their thinking processes, what Paul and Elder [66] refer to as thinking about thinking in order to improve thinking. This metacognitive dimension is crucial for developing critical thinking dispositions, as students reflect on and regulate their cognitive processes.

Another contributing factor was the flipped learning model's ability to allocate more classroom time for application-based activities [12, 16]. Debates, problem-solving exercises, and in-depth group discussions allowed students to structure their arguments more effectively and refine their ideas based on peer feedback.

Despite these positive outcomes, it is important to acknowledge that some studies have identified potential challenges in implementing flipped learning models. For instance, Bond [17] noted that while flipped learning generally enhances student engagement, factors such as lengthy videos and inadequate preparation can negatively impact student participation. Similarly, Akçayır and Akçayır , [2] highlighted issues related to student workload and the need for clear guidelines. Our study controlled for these potential challenges by carefully designing the AI-assisted flipped learning environment, including appropriately sized content chunks and clear instructional guidance, which may have contributed to the observed positive outcomes.

4.3 Effect on critical thinking motivation

The third key finding of this study is that students in the experimental group demonstrated more significant improvements in their motivation for engaging in critical thinking processes compared to those in the flipped-learning-only control group. The Critical Thinking Motivation Scale (CTM) measures students' general motivation toward critical thinking across multiple components, comprehensively assessing their overall motivational orientation toward critical thinking activities [83]. Our findings indicate significant improvements in students' total critical thinking motivation scores following the AI-assisted flipped learning intervention.

Existing research supports the idea that AI-assisted applications enhance student motivation [21, 42, 49, 75, 86]. Specifically, AI-driven platforms have been found to increase overall motivational engagement by offering personalized feedback and adaptive discussion environments. For example, Katona and Gyönyörű [49] demonstrated that integrating an AI-based adaptive learning system into flipped learning improved student engagement and motivation and created a more dynamic and collaborative learning experience. Similarly, Huang et al., [41] found that AI-assisted personalized video recommendations significantly enhanced students' motivation by increasing their engagement with learning materials.

This study aligns with previous research and highlights how the Socrat.ai platform contributes to developing critical thinking motivation. Its instant feedback mechanisms, individualized discussion scenarios, and dynamic content recommendations likely enhanced students' general motivation to engage in critical thinking activities. The AI-assisted environment increased students' overall appreciation for the importance and value of critical thinking processes by guiding students in structuring their thoughts and reinforcing their arguments with evidence. Crucially, because the control group also experienced flipped learning, with its known motivational benefits such as active in-class engagement and peer discussion, the motivational superiority of the experimental group ($t(68) = 3.189$, $p = .002$, $d = 0.76$) suggests that AI support may provide motivational benefits over and above those attributable to the flipped structure alone.

These findings can be understood through Valenzuela et al.,'s [83] application of Expectancy-Value Theory to critical thinking motivation, which suggests that students' engagement in critical thinking is influenced by their overall motivational orientation toward these cognitive processes. The AI-assisted flipped learning environment has positively impacted both components. By providing personalized feedback and creating opportunities to apply critical thinking skills successfully, the model likely enhanced students' overall confidence and perceived value of critical thinking activities.

The flipped learning model also supported critical thinking motivation, allowing for more in-class applications and group discussions [16, 82]. This structure enabled students to acquire fundamental knowledge before class and then apply it through complex problem-solving, analysis, and debate during class sessions.

However, some studies have identified challenges in maintaining student motivation in flipped learning environments. For instance, Debbağ and Yıldız [26] noted that while flipped learning can enhance motivation, technical issues can hinder student engagement. Similarly, Birgili and Demir , [14] found that students in flipped learning environments sometimes lack motivation, particularly for pre-class activities. Our findings suggest that AI assistance may be precisely the component that addresses this well-documented motivational challenge of flipped learning: by providing interactive, personalized pre-class engagement through Socrat.ai, the experimental group maintained higher motivational levels throughout the course compared to the control group, who completed pre-class preparation without AI support. Overall, the findings of this study suggest that integrating AI-assisted learning environments into flipped learning can lead to more significant and measurable improvements in students' overall critical thinking motivation.

4.4 Addressing the innovation effect

An important consideration in interpreting our findings is the potential impact of what is commonly known as the 'innovation effect' or 'novelty effect'—a phenomenon that refers to the temporary improvement in performance that may occur when participants are exposed to new technologies or teaching methods, primarily due to increased interest and engagement stemming from novelty rather than the inherent effectiveness of the intervention [81]. Given that AI-supported teaching models represent a relatively new and innovative approach, particularly in Turkish higher education, it is reasonable to question whether this effect influenced the observed improvements in critical thinking skills, dispositions, and motivation.

To address this concern, our study employed several methodological safeguards. First, the ten-week duration of the intervention allowed for the initial novelty to diminish, providing a more accurate assessment of the sustained impact of the AI-assisted flipped learning model. Research suggests that novelty effects typically wane after a few weeks of exposure to new technologies [67], and interventions lasting eight weeks or more are considered necessary for generalizable findings [22]. Second, validated assessment instruments measuring specific aspects of critical thinking rather than general engagement or satisfaction helped focus on substantive cognitive changes rather than affective responses to novelty. For instance, using the Istanbul Critical Thinking Scale (ICTS) ensured a focus on measurable cognitive outcomes, such as interpretation and evaluation, rather than subjective enthusiasm. Third, the active comparison group design, where the control group also experienced a contemporary, engaging pedagogy (flipped learning) rather than passive traditional instruction, ensured that the observed effects were not simply due to the novelty of any new approach, but specifically reflected the incremental value of AI assistance within a flipped learning context.

Nevertheless, we acknowledge that the novelty effect was not empirically tested in this study, which represents a limitation in isolating its potential influence on the observed outcomes. To better understand its impact, future research could employ experimental designs that directly measure the novelty effect by comparing student outcomes across groups exposed to AI-assisted learning for varying durations (e.g., short-term vs. long-term exposure) or by assessing engagement levels through qualitative measures like student interviews.

4.5 Practical implications

The findings of this study offer significant practical implications for educators and policymakers. By demonstrating that AI integration significantly enhances critical thinking outcomes even within an already-active flipped learning environment, the AAFL model suggests that instructors who already employ flipped learning can achieve substantially greater cognitive and motivational gains by incorporating AI tools, such as automated content recommendations, real-time feedback, and personalized learning environments, into their pre-class preparation phase.

Specifically, our findings suggest that instructors should integrate AI-assisted technologies that provide personalized feedback and adaptive content into their flipped classroom implementations. This approach can help address some of the challenges commonly associated with traditional flipped learning models, such as inadequate student preparation and limited real-time guidance [2, 14]. For example, educators can adopt platforms like Socrat.ai to deliver tailored pre-class materials, such as concise video lectures or interactive quizzes, that adapt to individual student needs, ensuring better preparation for in-class activities. Additionally, the results highlight the importance of balancing pre-class content delivery with in-class application activities that promote active engagement and collaborative learning, which is consistent with research indicating that active learning approaches enhance critical thinking development [15, 76].

To implement AAFL effectively, educators should follow these concrete steps: First, design pre-class content that is concise, engaging, and supported by AI-driven tools to provide immediate feedback, helping students identify gaps in their understanding

before class. Second, structure in-class sessions to include collaborative activities, such as small-group discussions or problem-solving tasks, that build on pre-class learning and encourage students to apply critical thinking skills like interpretation and evaluation. Third, AI-generated analytics can monitor student engagement and progress, allowing instructors to adjust content or provide targeted support during class. These strategies are particularly relevant for educators who have already adopted flipped learning but seek to amplify its effects on higher-order thinking through AI augmentation.

Educational institutions can optimize student-centered learning environments by integrating platforms like Socrat.ai, enhancing student engagement and accommodating individual learning differences. For instance, institutions can provide professional development workshops to train educators on using AI tools effectively, focusing on how to interpret AI-generated data to tailor instruction. Additionally, balancing pre-class preparation and in-class interaction can strengthen students' intrinsic motivation for learning. To sustain motivation, educators should incorporate AI-driven gamified elements, such as progress badges or personalized challenges, to make pre-class preparation more engaging. This approach enables educators and administrators to restructure teaching strategies and implement practical solutions for digital transformation in education.

4.6 Theoretical contribution

This study makes a significant theoretical contribution by providing one of the first controlled comparisons of AI-assisted flipped learning against flipped learning alone, offering a more precise analysis of how AI assistance, beyond the flipped structure itself, influences critical thinking development across three interconnected dimensions.

A key aspect of this study is its exploration of critical thinking processes across three interconnected dimensions. The cognitive dimension examines how students interpret, evaluate, and infer information, emphasizing their analytical depth. The dispositional dimension reflects students' openness to diverse perspectives and tendency to question claims critically, highlighting their willingness to engage in reflective skepticism. The motivational dimension considers both students' confidence in their ability to think critically and their recognition of its importance in academic and professional contexts.

Our findings contribute to existing theoretical frameworks by demonstrating how the AI component of AI-assisted flipped learning can simultaneously support multiple dimensions of critical thinking development. By integrating Vygotsky's (1978) social constructivism, Facione's (1990) conceptualization of critical thinking skills and dispositions, and Valenzuela et al.'s [83] application of Expectancy-Value Theory to critical thinking motivation, we offer a more comprehensive understanding of how AI-specific technological interventions can enhance critical thinking development within a flipped learning context. This integrated theoretical perspective provides a framework for understanding not only what critical thinking skills are developed but also why students might be more motivated to engage in critical thinking processes and how their dispositions toward critical thinking can be cultivated.

By analyzing these specific components, this study offers a more precise understanding of how AI-driven tools support critical thinking development. The personalized learning experiences and real-time feedback mechanisms provided by Socrat.ai have enabled a detailed assessment of improvements in critical thinking, extending beyond general learning outcomes. Theoretically, the study's active-control design offers a more

defensible causal account than designs that compare AI-assisted flipped learning against non-flipped traditional instruction.

Furthermore, our study extends Kong et al.,'s [52] findings on the efficacy of flipped classrooms in higher education by demonstrating that AI integration can enhance the already positive effects of flipped learning. This is particularly noteworthy because the control group in the present study already benefited from the well-documented advantages of flipped learning, yet still demonstrated significantly lower critical thinking outcomes than the AI-assisted group. This finding suggests that AI assistance represents a meaningful pedagogical addition rather than merely a repackaging of existing flipped learning benefits. Our findings indicate that AI assistance bridges pre-class content delivery and in-class application, providing the necessary scaffolding to ensure students are adequately prepared for higher-order thinking activities during class sessions.

4.7 Limitations

The redesigned study addresses the most fundamental limitation of the original design, the inability to disentangle the effects of AI assistance from those of flipped learning, by employing a control group that also received flipped learning without AI support. This design change substantially strengthens the causal claims regarding AI's contribution to critical thinking outcomes. Nevertheless, several limitations remain.

Although several covariates, such as gender and cumulative GPA, were included in sensitivity analyses to control for potential confounding variables, the influence of other unmeasured factors (e.g., students' learning styles, technology acceptance, and attitudes toward AI in education) remains unknown. Future research should incorporate these variables to understand better the mechanisms influencing critical thinking development. Relatedly, individual differences in students' epistemic beliefs, their assumptions about the nature and justification of knowledge, have been shown to shape how students engage in argumentation and reasoning tasks, including gendered patterns of engagement [10]. Because epistemic beliefs were not assessed in the present study, we cannot determine whether students with more sophisticated epistemic beliefs engaged more productively with Socrat.ai's Socratic prompts, or whether the AI-assisted condition was differentially effective across students with different baseline orientations toward knowledge and authority. This is a meaningful gap, given that 23 of 35 experimental-group participants were female; future studies should assess epistemic beliefs and AI literacy as potential moderators of gains in AI-assisted critical thinking. Additionally, some students encountered technical difficulties and adaptation challenges during the initial weeks of using the Socrat.ai platform. These issues may have influenced their engagement levels and learning outcomes, particularly during the early intervention phase. Ensuring more comprehensive technical orientation and support in future implementations may help mitigate such effects.

Another limitation concerns the potential novelty effect, whereby students' increased engagement may partly reflect the appeal of using new and interactive technology rather than the pedagogical value of the intervention itself. While the use of an active control group (flipped learning only) reduces, but does not eliminate, this concern, future research could further address it through longer-term implementations or staggered rollouts to differentiate between short-term enthusiasm and genuine pedagogical impact.

Two further design features constrain how specifically the observed gains can be attributed to AI per se. First, both groups were taught by the same instructor within sections of the same course; while this held instructional quality constant across conditions, the findings have not yet been replicated across instructors or course sections, and instructor-specific effects cannot be fully ruled out. Second, time on task during pre-class preparation was not measured or equated between groups: Socrat.ai's interactive, conversational format may have led experimental-group students to spend more time engaging with pre-class material than control-group students working with static digital materials. Because engagement time was not recorded, we cannot fully disentangle the effect of AI-specific features (e.g., adaptive questioning, longitudinal memory) from the more general effect of increased time on task. Future replications should log pre-class engagement time across conditions and involve multiple instructors to address these limitations. A further constraint on generalizability, distinct from the sample-size limitation discussed below, is that the study was conducted in a single elective course at one institution, with a self-selected population of students who chose to enroll in a critical-thinking course; the findings may not generalize to compulsory courses, other disciplines, or institutions with different student populations. Relatedly, although the AI-assisted condition was designed to differ from the control condition primarily in the presence of AI-generated questioning, feedback frequency, and the degree of accountability students felt for completing pre-class work, these factors may also have differed between conditions. For example, Socrat.ai's interactive format may have created a stronger sense of being "checked on" than the control group's static materials did. These factors were not measured or equated between groups and cannot be fully disentangled from the AI-specific features themselves.

A further limitation concerns the unexamined risk that AI-mediated dialogue could, under different conditions, undermine rather than support critical thinking. Although the present qualitative findings describe Socrat.ai as prompting deeper reasoning, a growing body of evidence cautions that generative AI use can also produce "metacognitive laziness", a tendency to offload cognitive effort onto the AI system rather than engage in independent reasoning, which has been associated with reduced learning gains when AI support is not carefully structured [34]. Similarly, Darvishi et al., [25] found that AI assistance can, in some implementations, diminish students' sense of agency over their learning. The present design, in which Socrat.ai's role was tightly scoped to pre-class Socratic questioning rather than answer generation, and in which in-class activities required students to apply their reasoning without AI support, was intended to mitigate this risk. However, the study did not directly measure cognitive offloading, overreliance, or agency. Future research should incorporate validated measures of these constructs to determine the conditions under which AI-assisted flipped learning supports, rather than substitutes for, independent critical thinking. Relatedly, students were not explicitly warned during orientation that Socrat.ai's questions or feedback could themselves be inaccurate or misleading. This omission may help explain the qualitative observation that students did not spontaneously question the substantive accuracy of the AI's prompts (see Qualitative Findings). Future implementations should include explicit guidance on the possibility of AI error as a safeguard against overreliance and unexamined trust in AI-generated content.

A related limitation concerns the relatively small sample size ($n = 70$, 35 per group). Although a priori power analysis confirmed adequate statistical power for the primary analyses, the modest sample limits the generalizability of findings to broader student populations and restricts the power available for subgroup analyses. Future research should seek to replicate these findings with larger, more diverse samples and across multiple institutions. Equally important, the ten-week intervention window and single post-test measurement point mean that this study can speak only to short-term gains; whether these improvements in critical thinking skills, dispositions, and motivation persist beyond the course or transfer to critical thinking tasks in other courses or to authentic, non-classroom problem-solving contexts remains untested. A delayed follow-up post-test and a transfer task administered outside the AAFL course would be needed to establish that the observed gains reflect durable cognitive change rather than short-term, context-bound performance. Additionally, although platform interaction data from Socrat.ai (including session duration and dialogue turn frequency) were available, these were not systematically integrated into the analysis. Relatedly, although Socrat.ai's question-generation logic is described in the Method, verbatim session-level transcripts were not archived for inclusion in this manuscript; future implementations of AI-assisted interventions should archive session logs to support full reproducibility. Future studies could benefit from learning analytics combined with extended mixed-methods approaches to further illuminate the mechanisms through which AI-assisted flipped learning supports critical thinking over time.

The present study took an initial step in this direction through semi-structured interviews with experimental-group students; however, this qualitative component has its own boundaries. Interviews were conducted only with experimental-group participants; without comparable qualitative data from the control group, we cannot determine whether the process-level experiences described here (e.g., engagement with pre-class preparation, preparedness for in-class discussion) are unique to the AI-assisted condition or are also present, to some degree, among students using non-AI flipped learning materials. The qualitative findings should therefore be read as an in-depth account of the experimental group's experience rather than as a between-group comparative analysis. In addition, the relatively small interview sample ($n = 9$) and single-institution context limit the generalisability of the qualitative findings. Future research would benefit from including comparative interviews with control-group participants, across multiple institutions.

Finally, the study's primary critical-thinking outcome measures were exclusively quantitative (ICTS, CTD, CTM). While these instruments offer objectivity and comparability, they may not fully capture the nuanced, process-oriented nature of critical thinking development. The study would have benefited from incorporating additional authentic assessment tasks, such as case analyses, research papers, or structured debates, that could demonstrate students' applied critical thinking abilities in more naturalistic contexts.

While the current design isolates AI's incremental contribution within a flipped learning context, it does not include an AI-only condition (without flipped learning) or a traditional instruction condition. A full multi-group design incorporating all four conditions, traditional instruction, flipped learning only, AI-assisted traditional instruction, and AI-assisted flipped learning, would permit a more complete decomposition of the

individual and interactive effects of each component, and is strongly recommended for future research.

5 Conclusion

This study provides evidence that AI-assisted flipped learning is associated with greater improvements in university students' critical thinking skills, dispositions, and motivation than flipped learning without AI support. By employing an active control group that also experienced flipped learning, the design offers more direct evidence than many prior studies that these gains are more plausibly attributable to the AI-assisted component than to the flipped structure alone, while acknowledging, as discussed in the Limitations, that factors such as differential time-on-task cannot be fully ruled out.

These findings suggest that educators who have already adopted flipped learning may find AI-assisted pre-class scaffolding, such as Socratic questioning and personalized feedback, a worthwhile addition for strengthening critical thinking outcomes. However, given the study's reliance on a single course, institution, and ten-week window, these implications should be treated as provisional. Future research involving multiple institutions, longer follow-up periods, and a fuller multi-group design (incorporating AI-only and non-flipped conditions) is needed to establish the generalisability and durability of these effects.

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E.Y. conceived the study, designed the research, collected and analyzed the data, and wrote the main manuscript. The author reviewed and approved the final version of the manuscript.

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The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
The study protocol was approved by the Bursa Uludağ University Social and Humanities Scientific Research Ethics Committee (Session No: 2024-09, Decision No: 18). The research was conducted in accordance with the ethical standards and guidelines of the committee and followed the principles outlined in the Higher Education Institutions Scientific Research and Publication Ethics Directive.

Informed consent
Informed consent to participate was obtained from all individual participants included in the study. Prior to data collection, participants were informed about the purpose and procedures of the study and their rights as participants. Participation was voluntary, and participants were assured that they could withdraw from the study at any time without any consequences.

Consent to publish
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References

1. Abrami PC, Bernard RM, Borokhovski E, Waddington DI, Wade CA, Persson T. Strategies for teaching students to think critically: A meta-analysis. Rev Educ Res. 2015;85(2):275–314. https://doi.org/10.3102/0034654314551063.
2. Akçayır G, Akçayır M. The flipped classroom: A review of its advantages and challenges. Comput Educ. 2018;126:334–45. https://doi.org/10.1016/j.compedu.2018.07.021.
3. Alan S, Yurt E. Flipped learning: An innovative model for enhancing education through ChatGPT. Int J Mod Educ Stud. 2024;8(1):124–48. https://doi.org/10.51383/ijonmes.2024.328.
4. Al-husban NA, Mashaleh M, Damrah L. Exploring first and fourth year university students' level of critical thinking skills at Arab Open University-Jordan. Curriculum Teach. 2022;37(2):55–72. https://doi.org/10.7459/ct/37.2.05.
5. Anderson LW, Krathwohl DR, Airasian P, Cruikshank K, Mayer R, Pintrich P, Wittrock M. A taxonomy for learning, teaching and assessing: A revision of Bloom's taxonomy. Longman Publishing; 2001.
6. Aykaç B, Aslan AE. Developing İstanbul critical thinking scale. J Hasan Ali Yücel Fac Educ. 2022;19(1):9–17. https://doi.org/10.5152/hayef.2021.21031.
7. Bachiller P, Badía G. The flip teaching as tool to improving students' sustainable learning performance in a financial course. Sustainability. 2020;12(23):9998. https://doi.org/10.3390/su12239998.
8. Baig MI, Yadegaridehkordi E. Flipped classroom in higher education: a systematic literature review and research challenges. Int J Educ Technol High Educ. 2023;20:61. https://doi.org/10.1186/s41239-023-00430-5.
9. Banihashem SK, Kerman NT, Noroozi O, Moon J, Drachsler H. Feedback sources in essay writing: peer-generated or AI-generated feedback? Int J Educational Technol High Educ. 2024;21(1):23. . https://doi.org/10.1186/s41239-024-00455-4
10. Banihashem SK, Noroozi O, Biemans HJA, Tassone VC. The intersection of epistemic beliefs and gender in argumentation performance. Innovations Educ Teach Int. 2024;61(4):716–34. . https://doi.org/10.1080/14703297.2023.2198995
11. Banihashem SK, Noroozi O, Khosravi H, Schunn CD, Drachsler H. Pedagogical framework for hybrid intelligent feedback. Innovations Educ Teach Int. 2025;1–17. https://doi.org/10.1080/14703297.2025.2499174.
12. Bergmann J, Sams A. Flip your classroom: Reach every student in every class every day. Eugene, OR: International Society for Technology in Education; 2012.
13. Bergmann J, Sams A. Flipped learning for social studies instruction. International Society for Technology in Education; 2015.
14. Birgili B, Demir Ö. An explanatory sequential mixed-method research on the full-scale implementation of flipped learning in the first years of the world's first fully flipped university: Departmental differences. Comput Educ. 2022;176:104352. https://doi.org/10.1016/j.compedu.2021.104352.
15. Birgili B, Seggie FN, Oğuz E. The trends and outcomes of flipped learning research between 2012 and 2018: A descriptive content analysis. J Comput Educ. 2021;8:365–94. https://doi.org/10.1007/s40692-021-00183-y.
16. Bishop JL, Verleger MA. (2013). The flipped classroom: A survey of the research. In *ASEE National Conference Proceedings* (pp. 1–18). https://doi.org/10.18260/1-2--22585
17. Bond M. Facilitating student engagement through the flipped learning approach in K-12: A systematic review. Comput Educ. 2020;151:103819. https://doi.org/10.1016/j.compedu.2020.103819.
18. Braun V, Clarke V. Using thematic analysis in psychology. Qualitative Res Psychol. 2006;3(2):77–101. https://doi.org/10.1191/1478088706qp063oa.
19. Brookfield SD, Preskill S. Discussion as a way of teaching: Tools and techniques for democratic classrooms. 2nd ed. Jossey-Bass; 2005.
20. Buil-Fabregá M, Martínez Casanovas M, Ruiz-Munzón N, Filho WL. Flipped classroom as an active learning methodology in sustainable development curricula. Sustainability. 2019;11(17):4577. https://doi.org/10.3390/su11174577.
21. Chiu MC, Hwang GJ, Hsia LH, Shyu FM. Artificial intelligence-supported art education: A deep learning-based system for promoting university students' artwork appreciation and painting outcomes. Interact Learn Environ. 2024;32(3):824–42. https://doi.org/10.1080/10494820.2022.2100426.
22. Chwo GSM, Marek MW, Wu WCV. Meta-analysis of MALL research and design. System. 2018;74:62–72. https://doi.org/10.1016/j.system.2018.02.009.
23. Cohen J. Statistical power analysis for the behavioral sciences. 2nd ed. Routledge; 1988.
24. Coursera. (2023). *Global skills report 2023*. Available online: https://downloads.ctfassets.net/2pudprfttvy6/3hsWnXIMBMffv3E8UiBrzC/fbe6c4a0a0dc3726e0825405c005f224/Coursera_Global_Skills_Report_2023.pdf (accessed on 09 May 2025).
25. Darvishi A, Khosravi H, Sadiq S, Gašević D, Siemens G. Impact of AI assistance on student agency. Comput Educ. 2024;210:104967. https://doi.org/10.1016/j.compedu.2023.104967.
26. Debbağ M, Yıldız S. Effect of the flipped classroom model on academic achievement and motivation in teacher education. Educ Inform Technol. 2021;26(3):3057–76. https://doi.org/10.1007/s10639-020-10395-x.
27. Deci EL, Ryan RM. The what and why of goal pursuits: Human needs and the self-determination of behavior. Psychol Inq. 2000;11(4):227–68. https://doi.org/10.1207/S15327965PLI1104_01.
28. Di Battista A, Grayling S, Hasselaar E, Leopold T, Li R, Rayner M, Zahidi S. (2023). *Future of jobs report 2023*. Geneva: World Economic Forum. Available online: https://www.weforum.org/publications/the-future-of-jobs-report-2023 (accessed on 10 May 2025).
29. Din M. Evaluating university students' critical thinking ability as reflected in their critical reading skill: A study at bachelor level in Pakistan. Think Skills Creativity. 2020;35:100627. https://doi.org/10.1016/j.tsc.2020.100627.
30. Downes S. (2012). *Connectivism and connective knowledge. Essays on meaning and learning networks*. My eBooks. Available online: http://www.downes.ca/me/mybooks.htm. (accessed on 09 May 2025).
31. Ennis RH. A logical basis for measuring critical thinking skills. Educational Leadersh. 1985;43(2):44–8.
32. Ennis RH. Critical thinking dispositions: their nature and assessability. Informal Log. 1996;18(2). https://doi.org/10.22329/il.v18i2.2378.
33. Facione PA. Critical thinking: A statement of expert consensus for purposes of educational assessment and instruction. California Academic; 1990. https://philarchive.org/archive/faccta.
34. Fan Y, Tang L, Le H, Shen K, Tan S, Zhao Y, Shen Y, Li X, Gašević D. Beware of metacognitive laziness: Effects of generative artificial intelligence on learning motivation, processes, and performance. Br J Edu Technol. 2025;56(2):489–530. https://doi.org/10.1111/bjet.13544.

35. Faul F, Erdfelder E, Lang AG, Buchner A. G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. Behav Res Methods. 2007;39:175–91. https://doi.org/10.3758/BF03193146.
36. Filali Marzouki O, Khalidi Idrissi M, Bennani S. Effects of social constructivist mobile learning environments on knowledge acquisition: A meta-analysis. Int J Interact Mob Technol (iJIM). 2017;11(1):18–39. https://doi.org/10.3991/ijim.v11i1.5982.
37. Fraga LM, Harmon J. The flipped classroom model of learning in higher education: An investigation of preservice teachers' perspectives and achievement. J Digit Learn Teacher Educ. 2015;31(1):18–27. https://doi.org/10.1080/21532974.2014.967420.
38. Gay LR, Mills GE, Airasian PW. Educational research: Competencies for analysts and applications. Harlow: Pearson; 2014.
39. Guilford JP. The nature of human intelligence. McGraw-Hill; 1967.
40. Hao Y. Exploring undergraduates' perspectives and flipped learning readiness in their flipped classrooms. Comput Hum Behav. 2016;59:82–92. https://doi.org/10.1016/j.chb.2016.01.032.
41. Huang AY, Lu OH, Yang SJ. Effects of artificial Intelligence–Enabled personalized recommendations on learners' learning engagement, motivation, and outcomes in a flipped classroom. Comput Educ. 2023;194:104684. . https://doi.org/10.1016/j.compedu.2022.104684
42. Huang X, Sun Y, Zhang L. Enhancing academic writing skills and motivation through AI-assisted learning. Front Psychol. 2023;14:1260843. . https://doi.org/10.3389/fpsyg.2023.1260843
43. Huber CR, Kuncel NR. Does College Teach Critical Thinking? A Meta-Analysis. Rev Educ Res. 2016;86(2):431–68. https://doi.org/10.3102/0034654315605917.
44. Huesca G, Martínez-Treviño Y, Molina-Espinosa JM, Sanromán-Calleros AR, Martínez-Román R, Cendejas-Castro EA, Bustos R. Effectiveness of using ChatGPT as a tool to strengthen benefits of the flipped learning strategy. Educ Sci. 2024;14(6):660. https://doi.org/10.3390/educsci14060660.
45. Järvelä S, Hadwin AF. New frontiers: Regulating learning in CSCL. Educational Psychol. 2013;48(1):25–39. https://doi.org/10.1080/00461520.2012.748006.
46. Jarvis P, Holford J, Griffin C. The theory and practice of learning. 2nd ed. Routledge; 2003.
47. Jintalan JR, Litao RA. Sustainability education and critical thinking integration in general education. Int J Curr Educ Stud (IJCES). 2024;3(2):51–76. https://doi.org/10.5281/zenodo.14544139.
48. Kaplan AM, Haenlein M. Rulers of the world, unite! The challenges and opportunities of artificial intelligence. Bus Horiz. 2019;62(1):37–50. https://doi.org/10.1016/j.bushor.2018.08.004.
49. Katona J, Gyonyoru KIK. Integrating AI-based adaptive learning into the flipped classroom model to enhance engagement and learning outcomes. Computers Education: Artif Intell. 2025;100392. https://doi.org/10.1016/j.caeai.2025.100392.
50. Kazanidis I, Pellas N, Fotaris P, Tsinakos A. Can the flipped classroom model improve students' academic performance and training satisfaction in Higher Education instructional media design courses? Br J Edu Technol. 2018;50(4):2014–27. https://doi.org/10.1111/bjet.12694.
51. Kong SC. An experience of a three-year study on the development of critical thinking skills in flipped secondary classrooms with pedagogical and technological support. Comput Educ. 2015;89:16–31. https://doi.org/10.1016/j.compedu.2015.08.017.
52. Kong W, Li D, Guo Q. Research on flipped classrooms in foreign language teaching in Chinese higher education. Humanit Social Sci Commun. 2024;11:525. https://doi.org/10.1057/s41599-024-03019-z.
53. Li H. Effects of a ChatGPT-based flipped learning guiding approach on learners' courseware project performances and perceptions. Australasian J Educational Technol. 2023;39(5):40–58. https://doi.org/10.14742/ajet.8923.
54. Li H, Cui Y. University students' cognitive flexibility and critical thinking dispositions. Front Psychol. 2025;15:1420272. https://doi.org/10.3389/fpsyg.2024.1420272.
55. Li Z, Li J. Using the flipped classroom to promote learner engagement for the sustainable development of language skills: A mixed-methods study. Sustainability. 2022;14(10):5983. https://doi.org/10.3390/su14105983.
56. Liu D, Zhang H. Improving students' higher order thinking skills and achievement using WeChat based flipped classroom in higher education. Educ Inform Technol. 2022;27(5):7281–302. https://doi.org/10.1007/s10639-022-10922-y.
57. Lodhi H, Noor S, Shafiq F. Exploring the progression of university students' critical thinking skills across academic years. Global Educational Stud Rev. 2023;VIIII:473–84. https://doi.org/10.31703/gesr.2023(VIII-I).41.
58. López-Villanueva D, Santiago R, Palau R. Flipped learning and artificial intelligence. Electronics. 2024;13(17):3424. https://doi.org/10.3390/electronics13173424.
59. Moore T. Critical thinking: Seven definitions in search of a concept. Stud High Educ. 2013;38(4):506–22. https://doi.org/10.1080/03075079.2011.586995.
60. Murillo-Zamorano LR, López Sánchez JÁ, Godoy-Caballero AL. How the flipped classroom affects knowledge, skills, and engagement in higher education: Effects on students' satisfaction. Computers Educ. 2019;141:103608. https://doi.org/10.1016/j.compedu.2019.103608.
61. Ng LK, Lo CK. Flipped classroom and gamification approach: Its impact on performance and academic commitment on sustainable learning in education. Sustainability. 2022;14(9):5428. https://doi.org/10.3390/su14095428.
62. Niu L, Behar-Horenstein LS, Garvan CW. Do instructional interventions influence college students' critical thinking skills? A meta-analysis. Educational Res Rev. 2013;9:114–28. https://doi.org/10.1016/j.edurev.2012.12.002.
63. OECD. *OECD Future of Education and Skills 2030*. OECD Learning Compass 2030. A Serie of Concept Notes. Paris: OECD; 2019.
64. OECD. (2023). *Future of education and skills 2030: Concept note*. OECD Publishing. Available online: https://www.oecd.org/content/dam/oecd/en/about/projects/edu/education-2040/concept-notes/Skills_for_2030_concept_note.pdf (accessed on 08 May 2025).
65. Orhan A. Comparing the effectiveness of online, flipped, and in-class critical thinking instruction on critical thinking skills and dispositions in higher education: Flipped classroom produces the greatest gains. Int J Technol Educ (IJTE). 2023;6(2):238–59. https://doi.org/10.46328/ijte.376.
66. Paul R, Elder L. Critical thinking: The nature of critical and creative thought. J Dev Educ. 2006;30(2):34–5. https://www.jstor.org/stable/42775621
67. Rodrigues L, Pereira FD, Toda AM, Palomino PT, Pessoa M, Carvalho LSG, Isotani S. Gamification suffers from the novelty effect but benefits from the familiarization effect: Findings from a longitudinal study. Int J Educ Technol High Educ. 2022;19:13. https://doi.org/10.1186/s41239-021-00314-6.

68. Rogers EM. Diffusion of innovations. 5th ed. Free; 2003.
69. Şanlı C. Artificial intelligence in geography teaching: potentialities, applications, and challenges. Int J Curr Educational Stud. 2025;4(1):47–76. https://doi.org/10.46328/ijces.170.
70. Schön DA. The reflective practitioner: How professionals think in action. Basic Books; 1983.
71. Sevillano-Monje V, Martín-Gutiérrez Á, Hervás-Gómez C. The flipped classroom and the development of competences: A teaching innovation experience in higher education. Educ Sci. 2022;12(4):248. https://doi.org/10.3390/educsci12040248.
72. Siemens G. Connectivism: A learning theory for the digital age. Int J Instructional Technol Distance Learn. 2005;2(1):3–10. https://jotamac.typepad.com/jotamacs_weblog/files/connectivism.pdf.
73. Sosu EM. The development and psychometric validation of a Critical Thinking Disposition Scale. Think Skills Creativity. 2013;9:107–19. https://doi.org/10.1016/j.tsc.2012.09.002.
74. Spielberger CD. Conceptual and methodological issues in anxiety research. In: Spielberger CD, editor. Anxiety: Current trends in theory and research. Volume 2. New York: Academic; 1972. pp. 481–93. https://doi.org/10.1016/B978-0-12-657402-9.50013-2.
75. Steen-Utheim AT, Foldnes N. A qualitative investigation of student engagement in a flipped classroom. Teach High Educ. 2017;23(3):307–24. https://doi.org/10.1080/13562517.2017.1379481.
76. Styers ML, Van Zandt PA, Hayden KL. Active learning in flipped life science courses promotes development of critical thinking skills. CBE—Life Sci Educ. 2018;17(3):ar39. https://doi.org/10.1187/cbe.16-11-0332.
77. Suwardika G, Sopandi AT, Indrawan IPO. (2024). AI-powered differentiated flipped classroom design thinking: fostering digital literacy, creative thinking, and self-efficacy. In *Proceeding of the International Conference on Innovation in Open and Distance Learning* (Vol. 5, pp. 574–625).
78. Tabachnick BG, Fidell LS. (2012). *Using multivariate statistics (6th ed.*). Pearson.
79. Torio H. Teaching as coaching: Experiences with a video-based flipped classroom combined with project-based approach in technology and physics higher education. J Technol Sci Educ. 2019;9(3):404–19. https://doi.org/10.3926/jotse.554.
80. Toulmin SE. (2003). *The uses of argument* (updated ed.). Cambridge University Press.
81. Tsay CH-H, Kofinas AK, Trivedi SK, Yang Y. Overcoming the novelty effect in online gamified learning systems: An empirical evaluation of student engagement and performance. J Comput Assist Learn. 2020;36:128–46. https://doi.org/10.1111/jcal.12385.
82. Tucker B. The flipped classroom. Educ Next. 2012;12(1):82–3. https://www.educationnext.org/wp-content/uploads/2023/04/ednext_XII_1_what_next.pdf.
83. Valenzuela J, Nieto A, Saiz C. Critical thinking motivational scale: A contribution to the study of relationship between critical thinking and motivation. Electron J Res Educ Psychol. 2011;9(24). https://doi.org/10.25115/ejrep.v9i24.1475.
84. Valenzuela J, Nieto A, Ossa C, Sepúlveda S, Muñoz C. Relationships between motivational factors and critical thinking. Eur J Educ Psychol. 2023;16(1):1–18. https://doi.org/10.32457/ejep.v16i1.2077.
85. Vygotsky LS. Mind in society: The development of higher psychological processes. Cambridge: Harvard University Press; 1978.
86. Yıldız E, Doğan U, Özbay Ö, Seferoğlu SS. Flipped classroom in higher education: An investigation of instructor perceptions through the lens of TPACK. Educ Inform Technol. 2022;27(8):10757–83. https://doi.org/10.1007/s10639-022-11059-8.
87. Yulian R. The flipped classroom: Improving critical thinking for critical reading of EFL learners in higher education. Stud Engl Lang Educ. 2021;8(2):508–22. https://doi.org/10.24815/siele.v8i2.18366.
88. Yurt E. The relationships among pre-service teachers' critical thinking disposition, self-efficacy, and creative thinking disposition in Turkey: A latent growth mediation model. Curr Psychol. 2025;44:85–102. https://doi.org/10.1007/s12144-024-07147-2.
89. Zou D, Xie H, Wang FL, Kwan R. Flipped learning with Wikipedia in higher education. Stud High Educ. 2020;45(5):1026–45. https://doi.org/10.1080/03075079.2020.1750195.

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